MODULE-3

1. Process Synchronization: Background

When multiple processes execute concurrently and access shared resources, synchronization ensures data consistency, avoiding conflicts and race conditions. It coordinates processes so that only one accesses critical resources at a time.

The Critical Section Problem

Consider a system consisting of n processes {Po, P1 , ... , P11 _ I}. Each process has a segment of code, called a cdticall in which the process may be changing common variables, updating a table, writing a file, and so on. The important feature of the system is that, when one process is executing in its critical section, no other process is to be allowed to execute in its critical section. That is, no two processes are executing in their critical sections at the same time. The critical-section problem is to design a protocol that the processes can use to cooperate. Each process must request permission to enter its critical section. The section of code implementing this request is the entire section. The critical section may be followed by an exit section. The remaining code is the reminder section. The general structure of a typical process Pi is shown in

```
do {
        entry section
            critical section
        exit section
            remainder section
} while (TRUE);
```

Figure 6.1 General structure of a typical process P_i.

Figure 6.1. The entry section and exit section are enclosed in boxes to highlight these important segments of code. A solution to the critical-section problem must satisfy the following three requirements:

1. Mutual exclusion. If process Pi is executing in its critical section, then no other processes can be executing in their critical sections.

2. Progress. If no process is executing in its critical section and some processes wish to enter their critical sections, then only those processes that are not executing in their remainder sections can participate in deciding which will enter its critical section next, and this selection carmot be postponed indefinitely.

3. Bounded waiting. There exists a bound, or limit, on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted.

We assume that each process is executing at a nonzero speed. However, we can make no assumption concerning the relative of the n processes. At a given point in time, many kernel-mode processes may be active in the operating system. As a result, the code implementing an operating system (kernel code) is subject to several possible race conditions. Consider as an example a kernel data structure that maintains a list of all open files in the system. This list must be modified when a new file is opened or closed (adding the file to the list or removing it from the list). If two processes were to open files simultaneously, the separate updates to this list could result in a race condition. Other kernel data structures that are prone to possible race conditions include structures for maintaining memory allocation, for maintaining process lists, and for interrupt handling. It is up to kernel developers to ensure that the operating system is free from such race conditions. Two general approaches are used to handle critical sections in operating systems: (1) preemptive kernels and (2) nonpreemptive kernels. A preemptive kernel allows a process to be preempted while it is running in kernel mode. A nonpreemptive kernel does not allow a process running in kernel mode to be preempted; a kernel-mode process will run until it exits kernel mode, blocks, or voluntarily yields control of the CPU. Obviously, a nonpreemptive kernel is essentially free from race conditions on kernel data structures, as only one process is active in the kernel at a time. We cannot say the same about preemptive kernels, so they must be carefully designed to ensure that shared kernel data are free from race conditions. Preemptive kernels are especially difficult to design for SMP architectures, since in these environments it is possible for two kernel-mode processes to run simultaneously on different processors.

- A critical section is a part of the code accessing shared resources.
- The problem is to ensure mutual exclusion: only one process can be in the critical section at once.
- Solutions must satisfy:
    - Mutual Exclusion
    - Progress (no indefinite postponement)
    - Bounded Waiting (limited waiting time for processes)

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2. Synchronization Mechanisms

- **Mutex Locks:** Provide mutual exclusion; a process must acquire the lock before entering its critical section, and release it after.one process at a time can only execute at a time.

  hese data structures are initialized to false. To prove that the mutualexclusion requirement is met, we note that process P; can enter its critical section only if either waiting [i] == false or key == false. The value of key can become false only if the TestAndSet () is executed. The first process to execute the TestAndSet () will find key== false; all others must wait. The variable waiting [i] can become false only if another process leaves its critical section; only one waiting [i] is set to false, maintaining the mutual-exclusion requirement.

```
do {
   key = TRUE;
   while (key == TRUE)
      Swap(&lock, &key);

      // critical section

   lock = FALSE;

      // remainder section
} while (TRUE);
```

Figure 6.7 Mutual-exclusion implementation with the Swap() instruction.

- **Semaphores:** The hardware-based solutions to the critical-section problem presented in Section 6.4 are complicated for application programmers to use. To overcmrte this difficulty, we can use a synchronization tool called a A semaphore S is an integer variable that, apart from initialization, is accessed only through two standard atomic operations: wait () and signal (). The wait () operation was originally termed P (from the Dutch proberen, "to test"); signal() was originally called V (from verhogen, "to increment"). The definition of wait () is as follows:

  Generalized locks that can control access to multiple instances of a resource. They use two atomic operations wait() (P) and signal() (V).

```
wait(S) {
    while S <= 0
       ; // no-op
    S--;
}
```

The definition of `signal()` is as follows:

```
signal(S) {
    S++;
}
```

All modifications to the integer value of the semaphore in the wait () and signal() operations must be executed indivisibly. That is, when one process modifies the semaphore value, no other process can simultaneously modify that same semaphore value. In addition, in the case of wait (S), the testing of the integer value of S (S :S 0), as well as its possible modification (S--), must be executed without interruption. We shall see how these operations can be implemented in Section 6.5.2; first, let us see how semaphores can be used.

6.5.2 Implementation

The main disadvantage of the semaphore definition given here is thatit requires While a process is in its critical section, any other process that tries to enter its critical section must loop continuously in the entry code. This continual looping is clearly a problem in a real multiprogramming system,

```
do {
   wait(mutex);

      // critical section

   signal(mutex);

      // remainder section
} while (TRUE);
```

Figure 6.9 Mutual-exclusion implementation with semaphores.

where a single CPU is shared among ncany processes. Busy waiting wastes CPU cycles that some other process might be able to use productively. This type of semaphore is also called a because the process "spins" while waiting for the lock. (Spinlocks do have an advantage in that no context switch is required when a process must wait on a lock, and a context switch may take considerable time. Thus, when locks are expected to be held for short times, spinlocks are useful; they are often employed on multiprocessor systems where one thread can "spin" on one processor while another thread performs its critical section on another processor.) To overcome the need for busy waiting, we can modify the definition of the wait() and signal() semaphore operations. When a process executes the wait () operation and finds that the semaphore value is not positive, it must wait. However, rather than engaging in busy waiting, the process can block itself. The block operation places a process into a waiting queue associated with the semaphore, and the state of the process is switched to the waiting state. Then control is transferred to the CPU scheduler, which selects another process to execute.

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3. Classic Synchronization Problems

- **<u>Producer-Consumer:</u>** Coordination of processes producing and consuming data with bounded buffer.

  The Bounded-Buffer Problem The bounded-buffer problem was introduced in Section 6.1; it is commonly used to illustrate the power of synchronization primitives. Here, we present a general structure of this scheme without committing ourselves to any particular implementation; we provide a related programming project in the exercises at the end of the chapter. We assume that the pool consists of n buffers, each capable of holding one item. The mutex semaphore provides mutual exclusion for accesses to the buffer pool and is initialized to the value 1. The empty and full semaphores comct the number of empty and full buffers. The semaphore empty is initialized to the value n; the semaphore full is initialized to the value 0. The code for the producer process is

shown in Figure 6.10; the code for the consumer process is shown in Figure 6.11. Note the symmetry between the producer and the consumer. We can interpret this code as the producer producing full buffers for the consumer or as the consumer producing empty buffers for the producer.

```
do {
   wait(full);
   wait(mutex);
      . . .
   // remove an item from buffer to nextc
      . . .
   signal(mutex);
   signal(empty);
      . . .
   // consume the item in nextc
      . . .
} while (TRUE);
```

Figure 6.11 The structure of the consumer process.

- **<u>Dining Philosophers:</u>** Avoid deadlock in a circular wait scenario.

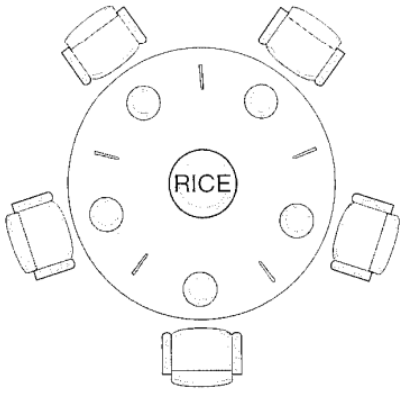

Figure 6.14 The situation of the dining philosophers.

to one philosopher. In the center of the table is a bowl of rice, and the table is laid with five single chopsticks (Figure 6.14). When a philosopher thinks, she does not interact with her colleagues. From time to time, a philosopher gets hungry and tries to pick up the two chopsticks that are closest to her (the chopsticks that are between her and her left and right neighbors). A

philosopher may pick up only one chopstick at a time. Obviously, she cam1ot pick up a chopstick that is already in the hand of a neighbor. When a htmgry philosopher has both her chopsticks at the same time, she eats without releasing her chopsticks. When she is finished eating, she puts down both of her chopsticks and starts thinking again. The dining-philosophers problem is considered a classic synchronization problem neither because of its practical importance nor because computer scientists dislike philosophers but because it is an example of a large class of concurrency-control problems. It is a simple representation of the need to allocate several resources among several processes in a deadlock-free and starvation-free mam1er. One simple solution is to represent each chopstick with a semaphore. A philosopher tries to grab a chopstick by executing await () operation on that semaphore; she releases her chopsticks by executing the signal() operation on the appropriate semaphores. Thus, the shared data are semaphore chopstick[5]; where all the elements of chopstick are initialized to 1. The structure of philosopher i is shown in Figure 6.15. Although this solution guarantees that no two neighbors are eating simultaneously, it nevertheless must be rejected because it could create a deadlock. Suppose that all five philosophers become hungry simultaneously and each grabs her left chopstick. All the elements of chopstick will now be equal to 0. When each philosopher tries to grab her right chopstick, she will be delayed forever.

- Several possible remedies to the deadlock problem are listed next. Allow at most four philosophers to be sitting simultaneously at the table.
- Allow a philosopher to pick up her chopsticks only if both chopsticks are available (to do this, she must pick them up in a critical section).
- Use an asymmetric solution; that is, an odd philosopher picks up first her left chopstick and then her right chopstick, whereas an even philosopher picks up her right chopstick and then her left chopstick

- **Readers-Writers:** Allow concurrent reads but exclusive writes to shared data.

  6.6.2 The Readers-Writers Problem Suppose that a database is to be shared among several concurrent processes. Some of these processes may want only to read the database, whereas others may want to update (that is, to read and write) the database. We distinguish between these two types of processes by referring to the former as readers and to the latter as writers. Obviously, if two readers access the shared data simultaneously, no adverse effects will result. However, if a writer and some other process (either a reader or a writer) access the database simultaneously, chaos may ensue. To ensure that these difficulties do not arise, we require that the writers have exclusive access to the shared database while writing to the database. This synchronization problem is referred to as the readers-writers problem. Since it was originally stated, it has been used to test nearly every new synchronization primitive. The readers-writers problem has several variations, all involving priorities. The simplest one, referred to as the first readers-writers problem, requires that no reader be kept waiting unless a writer has already obtained permission to use the shared object. In other words, no reader should wait for other readers to finish simply because a writer is waiting. The second readerswriters problem requires that, once a writer is ready, that writer performs its write as soon as possible. In other words, if a writer is waiting to

access the object, no new readers may start reading. A solution to either problem may result in starvation. In the first case, writers may starve; in the second case, readers may starve. For this reason, other variants of the problem have been proposed. Next, we present a solution to the first readers-writers problem. Refer to the bibliographical notes at the end of the chapter for references describing starvation-free solutions to the second readers-writers problem. In the solution to the first readers-writers problem, the reader processes share the following data structures: semaphore mutex, wrt; int readcount; The semaphores mutex and wrt are initialized to 1; readcount is initialized to 0. The semaphore wrt is common to both reader and writer processes. The mutex semaphore is used to ensure mutual exclusion when the variable readcount is updated. The readcount variable keeps track of how many processes are currently reading the object. The semaphore wrt functions as a mutual-exclusion semaphore for the writers. It is also used by the first or last reader that enters or exits the critical section. It is not used by readers who enter or exit while other readers are in their critical sections. The code for a writer process is shown in Figure 6.12; the code for a reader process is shown in Figure 6.13. Note that, if a writer is in the critical section and n readers are waiting, then one reader is queued on wrt, and n- 1 readers are queued on mutex. Also observe that, when a writer executes signal (wrt), we may resume the execution of either the waiting readers or a single waiting writer. The selection is made by the scheduler.

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4. Deadlocks

In a multiprogramming environment, several processes may compete for a finite number of resources. A process requests resources; if the resources are not available at that time, the process enters a waiting state. Sometimes, a waiting process is never again able to change state, because the resources it has requested are held by other waiting processes. This situation is called a deadlock.

System Model and Characterization

System Model

A system consists of a finite number of resources to be distributed among a number of competing processes. The resources are partitioned into several types, each consisting of some number of identical instances. Memory space, CPU cycles, files, and I/0 devices (such as printers and DVD drives) are examples of resource types. If a system has two CPUs, then the resource type CPU has two instances. Similarly, the resource type printer may have five instances. If a process requests an instance of a resource type, the allocation of any instance of the type will satisfy the request. If it will not, then the instances are not identical, and the resource type classes have not been defined properly. For example, a system may have two printers. These two printers may be defined to be in the same resource class if no one cares which printer prints which output. However, if one printer is on the ninth floor and the other is in the basement, then people on the ninth floor may not see both printers as equivalent, and separate resource classes may need to be defined for each printer. A process must request a resource before using it and must release the resource after using it. A process may request as

many resources as it requires to carry out its designated task. Obviously, the number of resources requested may not exceed the total number of resources available in the system. In other words, a process cannot request three printers if the system has only two. Under the normal mode of operation, a process may utilize a resource in only the following sequence:

- **Request.** The process requests the resource. If the request cannot be granted immediately (for example, if the resource is being used by another process), then the requesting process must wait until it can acquire the resource.
- **Use.** The process can operate on the resource (for example, if the resource is a printer, the process can print on the printer).
- **Release.** The process releases the resource.

Characterization

In a deadlock, processes never finish executing, and system resources are tied up, preventing other jobs from starting. Before we discuss the various methods for dealing with the deadlock problem, we look more closely at features that characterize deadlocks.

7.2.1 Necessary Conditions A deadlock situation can arise if the following four conditions hold simultaneously in a system:

Mutual exclusion. At least one resource must be held in a nonsharable mode; that is, only one process at a time can use the resource. If another DEADLOCK WITH MUTEX LOCKS Let's see how deadlock can occur in a multithreaded Pthread program using mutex locks. The pthread....mutex_ini t () function initializes an unlocked mutex. Mutex locks are acquired and released using pthread....mutex_lock() and pthread....mutex_unlock (), respectively. If a thread attempts to acquire a locked mutex, the call to pthread....mutex_lock 0 blocks the thread until the owner of the mutex lock invokes pthread....mutex_unlock (). Two mutex locks are created in the following code example: I* Create and initialize the mutex locks *I pthread....mutex_t first....mutex; pthread....mutex_t second_nmtex; pthread....mutex_ini t (&first....mutex, NULL) ; pthread....mutex_ini t (&second....mutex, NULL) ; Next, two threads-thread_one and thread_two-'-are created, and both these threads have access to both mutex locks. thread_one and thread_ two run in the functions do_work_one () and do_work_two (), respectively, as shown in Figure 7.1.

- Deadlock happens when processes wait indefinitely for resources held by each other.
- Characterized by four conditions:
    1. **Mutual Exclusion**
    2. **Hold and Wait:-** Hold and wait. A process must be holding at least one resource and waiting to acquire additional resources that are cmrently being held by other processes.
    3. **No Pre-emption :-**Resources cannot be preempted; that is, a resource can be released only voluntarily by the process holding it, after that process has completed its task.

4. **Circular Wait:-** Circular wait. A set { P0 , Pl, ... , P11 } of waiting processes must exist such that Po is waiting for a resource held by P1, P1 is waiting for a resource held by P2, ... , Pn-1 is waiting for a resource held by P,v and P11 is waiting for a resource held by Po.

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5. Methods for Handling Deadlocks

Generally speaking, we can deal with the deadlock problem in one of three ways:

- **We can use a protocol to prevent or avoid deadlocks, ensuring that the system will never enter a deadlocked state.**
- **We can allow the system to enter a deadlocked state, detect it, and recover.**
- **We can ignore the problem altogether and pretend that deadlocks never occur in the system. The third solution is the one used by most operating systems, including UNIX and Windows; it is then up to the application developer to write programs that handle deadlocks.**

1. **Deadlock Prevention**: Ensure at least one of the necessary conditions for deadlock cannot hold.

   As we noted in Section 7.2.1, for a deadlock to occur, each of the four necessary conditions must hold. By ensuring that at least one of these conditions cannot hold, we can prevent the occurrence of a deadlock. We elaborate on this approach by examining each of the four necessary conditions separately.

   7.4.1 Mutual Exclusion The mutual-exclusion condition must hold for nonsharable resources. For example, a printer cannot be simultaneously shared by several processes. Sharable resources, in contrast, do not require mutually exclusive access and thus cannot be involved in a deadlock. Read-only files are a good example of a sharable resource. If several processes attempt to open a read-only file at the same time, they can be granted simultaneous access to the file. A process never needs to wait for a sharable resource. In general, however, we cannot prevent deadlocks by denying the mutual-exclusion condition, because some resources are intrinsically nonsharable.

   7.4.2 Hold and Wait To ensure that the hold-and-wait condition never occurs in the system, we must guarantee that, whenever a process requests a resource, it does not hold any other resources. One protocol that can be used requires each process to request and be allocated all its resources before it begins execution. We can implement this provision by requiring that system calls requesting resources for a process precede all other system calls. An alternative protocol allows a process to request resources only when it has none. A process may request some resources and use them. Before it can request any additional resources, however, it must release all the resources that it is currently allocated. To illustrate the difference between these two protocols, we consider a process that copies data from a DVD drive to a file on disk, sorts the file, and then prints the results to a printer. If all resources must be requested at the beginning

of the process, then the process must initially request the DVD drive, disk file, and printer. It will hold the printer for its entire execution, even though it needs the printer only at the end. The second method allows the process to request initially only the DVD drive and disk file. It copies from the DVD drive to the disk and then releases 292 Chapter 7 both the DVD drive and the disk file. The process must then again request the disk file and the printer. After copying the disk file to the printer, it releases these two resources and terminates. Both these protocols have two main disadvantages. First, resource utilization may be low, since resources may be allocated but unused for a long period. In the example given, for instance, we can release the DVD drive and disk file, and then again request the disk file and printe1~ only if we can be sure that our data will remain on the disk file. Otherwise, we must request all resources at the beginning for both protocols. Second, starvation is possible. A process that needs several popular resources may have to wait indefinitely, because at least one of the resources that it needs is always allocated to some other process.

7.4.3 No Preemption The third necessary condition for deadlocks is that there be no preemption of resources that have already been allocated. To ensure that this condition does not hold, we can use the following protocol. If a process is holding some resources and requests another resource that cannot be immediately allocated to it (that is, the process must wait), then all resources the process is currently holding are preempted. In other words, these resources are implicitly released. The preempted resources are added to the list of resources for which the process is waiting. The process will be restarted only when it can regain its old resources, as well as the new ones that it is requesting. Alternatively, if a process requests some resources, we first check whether they are available. If they are, we allocate them. If they are not, we check whether they are allocated to some other process that is waiting for additional resources. If so, we preempt the desired resources from the waiting process and allocate them to the requesting process. If the resources are neither available nor held by a waiting process, the requesting process must wait. While it is waiting, some of its resources may be preempted, but only if another process requests them. A process can be restarted only when it is allocated the new resources it is requesting and recovers any resources that were preempted while it was waiting. This protocol is often applied to resources whose state can be easily saved and restored later, such as CPU registers and memory space. It cannot generally be applied to such resources as printers and tape drives.

7 .4.4 Circular Wait The fourth and final condition for deadlocks is the circular-wait condition. One way to ensure that this condition never holds is to impose a total ordering of all resource types and to require that each process requests resources in an increasing order of enumeration. To illustrate, we let R = { R1, R2, ... , Rm} be the set of resource types. We assign to each resource type a unique integer number, which allows us to compare two resources and to determine whether one precedes another in our ordering. Formally, we define a one-to-one hmction F: R ___,. N, where N is the set of natural numbers. For example, if the set of resource types R includes tape drives, disk drives, and printers, then the function F might be defined as follows:

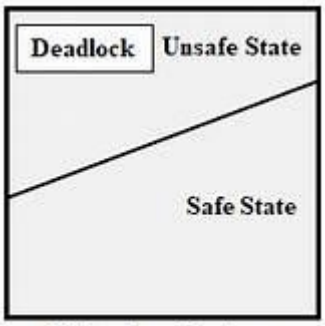
We can now consider the following protocol to prevent deadlocks: Each process can request resources only in an increasing order of enumeration. That is, a process can initially request any number of instances of a resource type -say, R;. After that, the process can request instances of resource type Rj if and only if F(Rj) > F(R;). For example, using the function defined previously, a process that wants to use the tape drive and printer at the same time must first request the tape drive and then request the printer. Alternatively, we can require that a process requesting an instance of resource type Rj must have released any resources R; such that F(Ri) ::=:: F(Rj). It must also be noted that if several iilstances of the same resource type are needed, a single request for all of them must be issued. If these two protocols are used, then the circular-wait condition cannot hold. We can demonstrate this fact by assuming that a circular wait exists (proof by contradiction). Let the set of processes involved in the circular wait be { P0 , P1, ... , P11 }, where Pi is waiting for a resource R;, which is held by process Pi+l· (Modulo arithmetic is used on the indexes, so that P11 is waiting for a resource R11 held by P0 .) Then, since process Pi+l is holding resource Ri while requesting resource Ri+l' we must have F(Ri) < F(R;H) for all i. But this condition means that F(Ro) < F(R1) < ... < F(R11) < F (Ro). By transitivity, F(Ro) < F(Ro), which is impossible. Therefore, there can be no circular wait. We can accomplish this scheme in an application program by developing an ordering among all synchronization objects in the system. All requests for synchronization objects must be made in increasing order. For example, if the lock ordering in the Pthread program shown in Figure 7.1 was F (first_mutex) = 1 F (second_mutex) = 5

2. **Deadlock Avoidance**: Requires additional information about future resource requests (e.g., Banker's Algorithm).

Deadlock-prevention algorithms, as discussed in Section 7.4, prevent deadlocks by restraining how requests can be made. The restraints ensure that at least one of the necessary conditions for deadlock cannot occur and, hence, that deadlocks cannot hold. Possible side effects of preventing deadlocks by this method, however, are low device utilization and reduced system throughput. An alternative method for avoiding deadlocks is to require additional information about how resources are to be requested. For example, in a system with one tape drive and one printer, the system might need to know that process P will request first the tape drive and then the printer before releasing both resources, whereas process Q will request first the printer and then the tape drive. With this knowledge of the complete sequence of requests and releases for each process, the system can decide for each request whether or not the process should wait in order to avoid a possible future deadlock. Each request requires that in making this decision the system consider the resources currently available, the resources currently allocated to each process, and the future requests and releases of each process.

7.5.1 Safe State A state is safe if the system can allocate resources to each process (up to its maximum) in some order and still avoid a deadlock. More formally, a system is in a safe state only if there exists a safe sequence. A sequence of processes is a safe sequence for the current allocation state if, for each Pi, the resource requests that Pi can

still make can be satisfied by the currently available resources plus the resources held by all Pj, with j < i. In this situation, if the resources that Pi needs are not immediately available, then Pi can wait until all Pj have finished. When they have finished, Pi can obtain all of its needed resources, complete its designated task, return its allocated resources, and terminate. When Pi terminates, Pi+l can obtain its needed resources, and so on. If no such sequence exists, then the system state is said to be unsafe. A safe state is not a deadlocked state. Conversely, a deadlocked state is an unsafe state. Not all unsafe states are deadlocks, however (Figure 7.5). An unsafe state may lead to a deadlock. As long as the state is safe, the operating system can avoid unsafe (and deadlocked) states. In an unsafe state, the operating system cannot prevent processes from requesting resources in such a way that a deadlock occurs. The behavior of the processes controls unsafe states.

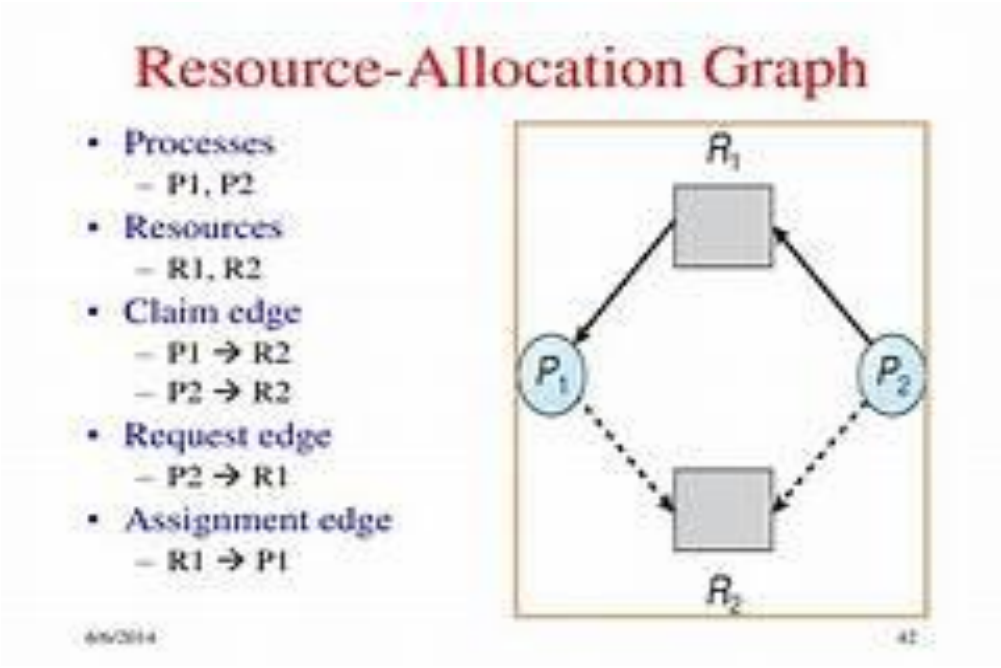


States in a System

7.5.2 Resource-Allocation-Graph Algorithm

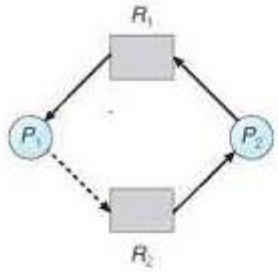
If we have a resource-allocation system with only one instance of each resource type, we can use a variant of the resource-allocation graph defined in Section 7.2.2 for deadlock avoidance. In addition to the request and assignment edges already described, we introduce a new type of edge, called a claim edge.

A claim edge Pi ~ Rj indicates that process Pi may request resource Rj at some time in the future. This edge resembles a request edge in direction but is represented in the graph by a dashed line. When process Pi requests resource



R1, the claim edge P; -+ R1 is converted to a request edge. Similarly, when a resource R1 is released by P;, the assignment edge Rj -+ P; is reconverted to a claim edge P; -+ Rj. We note that the resources must be claimed a priori in the system. That is, before process P; starts executing, all its claim edges must already appear in the resource-allocation graph. We can relax this condition by allowing a claim edge P; -+ R1 to be added to the graph only if all the edges associated with process P; are claim edges. Now suppose that process P; requests resource Rj. The request can be granted only if converting the request edge P; -+ Rj to an assignment edge R1 -+ P; does not result in the formation of a cycle in the resource-allocation graph. We check for safety by using a cycle-detection algorithm. An algorithm for detecting a cycle in this graph requires an order of n2 operations, where n is the number of processes in the system. If no cycle exists, then the allocation of the resource will leave the system in a safe state. If a cycle is found, then the allocation will put the system in an unsafe state. In that case, process P; will have to wait for its requests to be satisfied. To illustrate this algorithm, we consider the resource-allocation graph of Figure 7.6. Suppose that P2 requests R2 . Although R2 is currently free, we cannot allocate it to P2, since this action will create a cycle in the graph (Figure 7.7). A cycle, as mentioned, indicates that the system is in an unsafe state. If P1 requests R2, and P2 requests R1, then a deadlock will occur.

Unsafe State In Resource-Allocation Graph



7.5.3 Banker's Algorithm The resource-allocation-graph algorithm is not applicable to a resourceallocation system with multiple instances of each resource type. The deadlockavoidance algorithm that we describe next is applicable to such a system but is less efficient than the resource-allocation graph scheme. This algorithm is commonly known as the banker's algorithm. The name was chosen because the algorithm. could be used in a banking system to ensure that the bank never allocated its available cash in such a way that it could no longer satisfy the needs of all its customers. When a new process enters the system, it must declare the maximum number of instances of each resource type that it may need. This nun1.ber may not exceed the total number of resources in the system. When a user requests a set of resources, the system must determine whether the allocation of these resources will leave the system in a safe state. If it will, the resources are allocated; otherwise, the process must wait until some other process releases enough resources. Several data structures must be maintained to implement the banker's algorithm. These data structures encode the state of the resource-allocation system. We need the following data structures, where n is the number of processes in the system and m is the number of resource types:

Available. A vector of length m indicates the number of available resources of each type. If Available[j] equals k, then k instances of resource type Ri are available.

Max. An n x m matrix defines the maximum demand of each process. If Max[i] [j] equals k, then process P; may request at most k instances of resource type Ri.

Allocation. An 11 x m matrix defines the number of resources of each type currently allocated to each process. If Allocation[i][j] equals lc, then process P; is currently allocated lc instances of resource type Rj.

Need. An n x m matrix indicates the remaining resource need of each process. If Need[i][j] equals k, then process P; may need k more instances of resource type Ri to complete its task. Note that Need[i][j] equals Max[i][j] - Allocation [i][j].

7.5.3.2 Resource-Request Algorithm

Next, we describe the algorithm for determining whether requests can be safely granted. Let Request; be the request vector for process P;. If Request; [j] == k, then process P; wants k instances of resource type Rj. When a request for resources is made by process P;, the following actions are taken: If Request; ::::; Need;, go to step 2. Otherwise, raise an error condition, since the process has exceeded its maximum claim. If Request; ::; Available, go to step 3. Otherwise, P; must wait, since the resources are not available.

Have the system pretend to have allocated the requested resources to process P; by modifyil1.g the state as follows: Available= Available- Request;; Allocation; =Allocation; +Request;; Need; =Need;- Request;;

7.5.3.3 An Illustrative Example To illustrate the use of the banker's algorithm, consider a system with five processes Po through P4 and three resource types A, B, and C. Resource type A has ten instances, resource type B has five instances, and resource type C has seven instances. Suppose that, at time T0 , the following snapshot of the system has been taken:

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

The content of the matrix *Need* is defined to be *Max* − *Allocation* and is as follows:

	Need
	A B C
P_0	7 4 3
P_1	1 2 2
P_2	6 0 0
P_3	0 1 1
P_4	4 3 1

We claim that the system is currently in a safe state. Indeed, the sequence $<P_1, P_3, P_4, P_2, P_0>$ satisfies the safety criteria. Suppose now that process P_1 requests one additional instance of resource type *A* and two instances of resource type *C*, so $Request_1 = (1,0,2)$. To decide whether this request can be immediately granted, we first check that $Request_1 \leq Available$—that is, that $(1,0,2) \leq (3,3,2)$, which is true. We then pretend that this request has been fulfilled, and we arrive at the following new state:

	Allocation	*Need*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 4 3	2 3 0
P_1	3 0 2	0 2 0	
P_2	3 0 2	6 0 0	
P_3	2 1 1	0 1 1	
P_4	0 0 2	4 3 1	

We must determine whether this new system state is safe. To do so, we execute our safety algorithm and find that the sequence $<P_1, P_3, P_4, P_0, P_2>$ satisfies the safety requirement. Hence, we can immediately grant the request of process P_1.

You should be able to see, however, that when the system is in this state, a request for (3,3,0) by P_4 cannot be granted, since the resources are not available. Furthermore, a request for (0,2,0) by P_0 cannot be granted, even though the resources are available, since the resulting state is unsafe.

We leave it as a programming exercise for students to implement the banker's algorithm.

3. **<u>Deadlock Detection and Recovery</u>:** Detect circular waits and recover by terminating or rolling back processes.

7.7.1 Process Termination To eliminate deadlocks by aborting a process, we use one of two methods. In both methods, the system reclaims all resources allocated to the terminated processes

Abort all deadlocked processes. This method clearly will break the deadlock cycle, but at great expense; the deadlocked processes may have computed for a long time, and the results of these partial computations must be discarded and probably will have to be recomputed later.

Abort one process at a time until the deadlock cycle is eliminated. This method incurs considerable overhead, since after each process is aborted, a deadlock-detection algorithnc rnust be invoked to determine whether any processes are still deadlocked.

7.7.2 Resource Preemption To eliminate deadlocks using resource preemption, we successively preempt some resources from processes and give these resources to other processes 1-m til the deadlock cycle is broken. If preemption is required to deal with deadlocks, then three issues need to be addressed:

- **Selecting a victim,**
- **Rollback,**
- **Starvation.**

- Resource preemption is a deadlock recovery technique in which the operating system temporarily takes resources away from one or more processes and reallocates them to other processes until the deadlock cycle is broken. When a deadlock occurs, simply waiting for processes to release resources on their own is not possible, so the system must intervene by forcibly preempting resources. Although this approach can effectively resolve deadlocks, it introduces several practical challenges that must be carefully handled.
- The first issue is **selecting a victim**. When resources need to be preempted, the operating system must decide which process should lose its resources. This decision is important because preempting a process may cause loss of work and reduce system performance. To minimize the negative impact, the system selects a victim based on factors such as process priority, the number and type of resources it holds, how much execution time it has already used, and how much time it still needs to complete. The goal is to choose a process whose preemption causes the least overall cost to the system.
- The second issue is **rollback**. Once a process is selected as a victim and its resources are taken away, it cannot continue execution from its current state. The operating system must roll the process back to a previously saved safe state, known as a checkpoint, and restart its execution from there. This ensures system consistency and correctness after resource preemption. However, maintaining checkpoints and rolling back processes adds overhead, and some operations, especially input/output operations, cannot be easily undone.

- The third issue is **starvation**. Starvation occurs when the same process is repeatedly chosen as the victim and is continuously rolled back, preventing it from ever completing its execution. This usually happens when low-priority processes are always selected for preemption. To prevent starvation, the operating system may limit the number of times a process can be preempted or gradually increase its priority after each rollback, ensuring that every process eventually gets a fair chance to execute and complete.

MODULE-4

1. Memory Management Strategies: Background

Memory management controls allocation and deallocation of memory to processes, ensuring efficient use, isolation, and preventing conflicts. It addresses issues like fragmentation, protection, and sharing of memory.

As we saw in Chapter 1, memory is central to the operation of a modern computer system. Memory consists of a large array of words or bytes, each with its own address. The CPU fetches instructions from memory according to the value of the program counter. These instructions may cause additional loading from and storing to specific memory addresses. A typical instruction-execution cycle, for example, first fetches an instruction from memory. The instruction is then decoded and may cause operands to be fetched from memory. After the instruction has been executed on the

operands, results may be stored back in memory. The mernory unit sees only a stream of memory addresses; it does not know how they are generated (by the instruction counter, indexing, indirection, literal addresses, and so on) or what they are for (instructions or data). Accordingly, we can ignore hozu a program generates a memory address. We are interested only in the sequence of memory addresses generated by the running program. We begin our discussion by covering several issues that are pertinent to the various techniques for managing memory. This coverage includes an overview of basic hardware issues, the binding of symbolic memory addresses to actual physical addresses, and the distinction between logical and physical addresses. We conclude the section with a discussion of dynamically loading and linking code and shared libraries.

8.1.1 Basic Hardware Main memory and the registers built into the processor itself are the only storage that the CPU can access directly. There are machine instructions that take memory addresses as arguments, but none that take disk addresses. Therefore, any instructions in execution, and any data being used by the instructions, must be in one of these direct-access storage devices. If the data are not in memory, they must be moved there before the CPU can operate on them. Registers that are built into the CPU are generally accessible within one cycle of the CPU clock. Most CPUs can decode instructions and perform simple operations on register contents at the rate of one or more operations per clock tick The same cannot be said of main memory, which is accessed via a transaction on the memory bus. Completing a memory access may take many cycles of the CPU clock. In such cases, the processor normally needs to stall, since it does not have the data required to complete the instruction that it is executing. This situation is intolerable because of the frequency of memory accesses. The remedy is to add fast memory between the CPU and

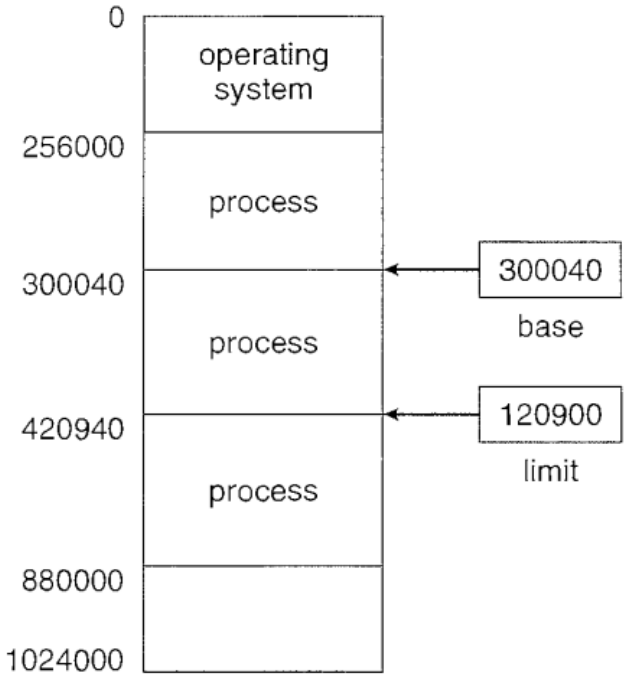


Figure 8.1 A base and a limit register define a logical address space.

main memory. A memory buffer used to accommodate a speed differential, called a cache, is described in Section 1.8.3. Not only are we concerned with the relative speed of accessing physical memory, but we also must ensure correct operation to protect the operating system from access by user processes and, in addition, to protect user processes from one another. This protection must be provided by the hardware. It can be implemented in several ways, as we shall see throughout the chapter. In this section, we outline one possible implementation. We first need to make sure that each process has a separate memory space. To do this, we need the ability to determine the range of legal addresses that the process may access and to ensure that the process can access only these legal addresses. We can provide this protection by using two registers, usually a base and a limit, as illustrated in Figure 8.1. The base register holds the smallest legal physical memory address; the limit register specifies the size of the range. For example, if the base register holds 300040 and the limit register is 120900, then the program can legally access all addresses from 300040 through 420939 (inclusive). Protection of memory space is accomplished by having the CPU hardware compare every address generated in user mode with the registers. Any attempt by a program executing in user mode to access operating-system memory or other users' memory results in a trap to the operating system, which treats the attempt as a fatal error (Figure 8.2). This scheme prevents a user program from (accidentally or deliberately) modifying the code or data structures of either the operating system or other users. The base and limit registers can be loaded only by the operating system, which uses a special privileged instruction. Since privileged instructions can be executed only in kernel mode, and since only the operating system executes in kernel mode, only the operating system can load the base and limit registers. This scheme allows the operating system to change the value of the registers but prevents user programs from changing the registers' contents. The operating system, executing in kernel mode, is given unrestricted access to both operating system memory and users' memory. This provision allows the operating system to load users'

programs into users' memory, to

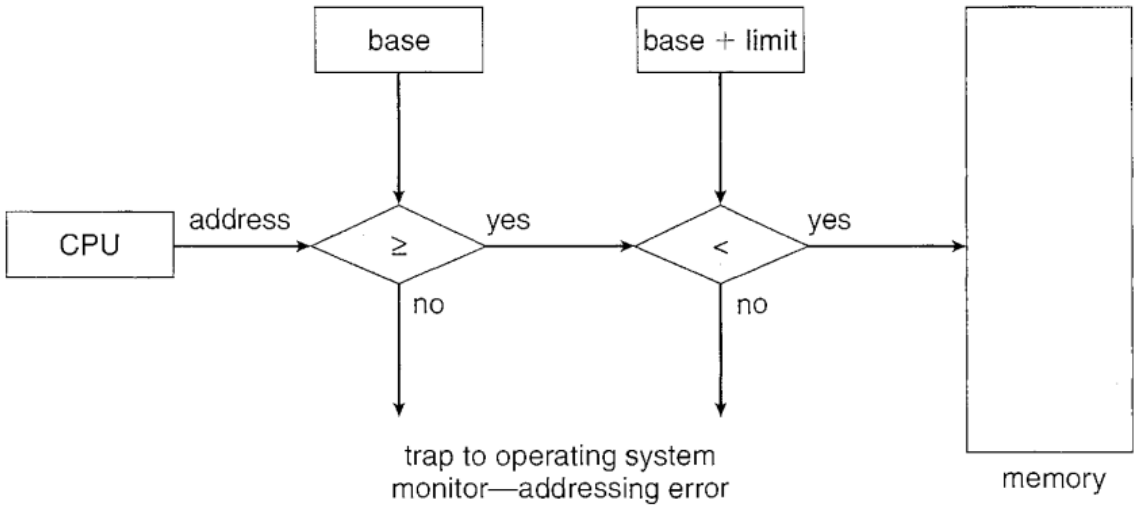


Figure 8.2 Hardware address protection with base and limit registers.

8.1.2 Address Binding Usually

a program resides on a disk as a binary executable file. To be executed, the program must be brought into memory and placed within a process. Depending on the memory management in use, the process may be moved between disk and memory during its execution. The processes on the disk that are waiting to be brought into memory for execution form the The normal procedure is to select one of the processes in the input queue and to load that process into memory. As the process is executed, it accesses instructions and data from memory. Eventually, the process terminates, and its memory space is declared available. Most systems allow a user process to reside in any part of the physical memory. Thus, although the address space of the computer starts at 00000, the first address of the user process need not be 00000. This approach affects the addresses that the user program can use. In most cases, a user program will go through several steps-some of which may be optional-before bein.g executed (Figure 8.3). Addresses may be represented in different ways during these steps. Addresses in the source program are generally symbolic (such as count). A compiler will typically bind these symbolic addresses to relocatable addresses (such as "14 bytes from the beginning of this module"). The lin.kage editor or loader will in turn bind the relocatable addresses to absolute addresses (such as 74014). Each binding is a mapping from one address space to another. Classically, the binding of instructions and data to memory addresses can be done at any step along the way:

- **Compile time.** If you know at compile time where the process will reside in memory, then Absolut code can be generated. For example, if you krww that a user process will reside starting at location R, then the generated compiler code will start at that location and extend up from there. If, at some later time, the starting location changes, then it will be necessary to recompile this code. The MS-DOS .COM-format programs are bound at compile time.
- **Load time.** If it is not known at compile time where the process will reside in memory, then the compiler must generate relocatable code. In this case, final binding is delayed until load time. If the starting address changes, we need only reload the user code to incorporate this changed value.
- **Execution time.** If the process can be moved during its execution from one memory segment to another, then binding must be delayed until run time. Special hardware must

be available for this scheme to work, as will be discussed in Section 8.1.3. Most general-purpose operating systems 11se this method.

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2. Swapping

Swapping moves processes between main memory and a backing store (disk) to free memory temporarily. This allows the CPU to execute processes partially loaded into memory, improving CPU utilization.

A process must be in memory to be executed. A process, however, can be swapped temporarily out of memory to a backing store and then brought into memory for continued execution. For example, assume a multiprogramming environment with a round-robin CPU-scheduling algorithm. When a quantum expires, the memory manager will start to swap out the process that just finished and to swap another process into the memory space that has been freed (Figure 8.5). In the meantime, the CPU scheduler will allocate a time slice to some other process in memory. When each process finishes its quantum, it will be swapped with another process. Ideally, the memory manager can swap processes fast enough that some processes will be in memory, ready to execute, when the CPU scheduler wants to reschedule the CPU. In addition, the quantum must be large enough to allow reasonable amounts of computing to be done between swaps. A variant of this swapping policy is used for priority-based scheduling algorithms. If a higher-priority process arrives and wants service, the memory manager can swap out the lower-priority process and then load and execute the higher-priority process. When the higher-priority process finishes, the

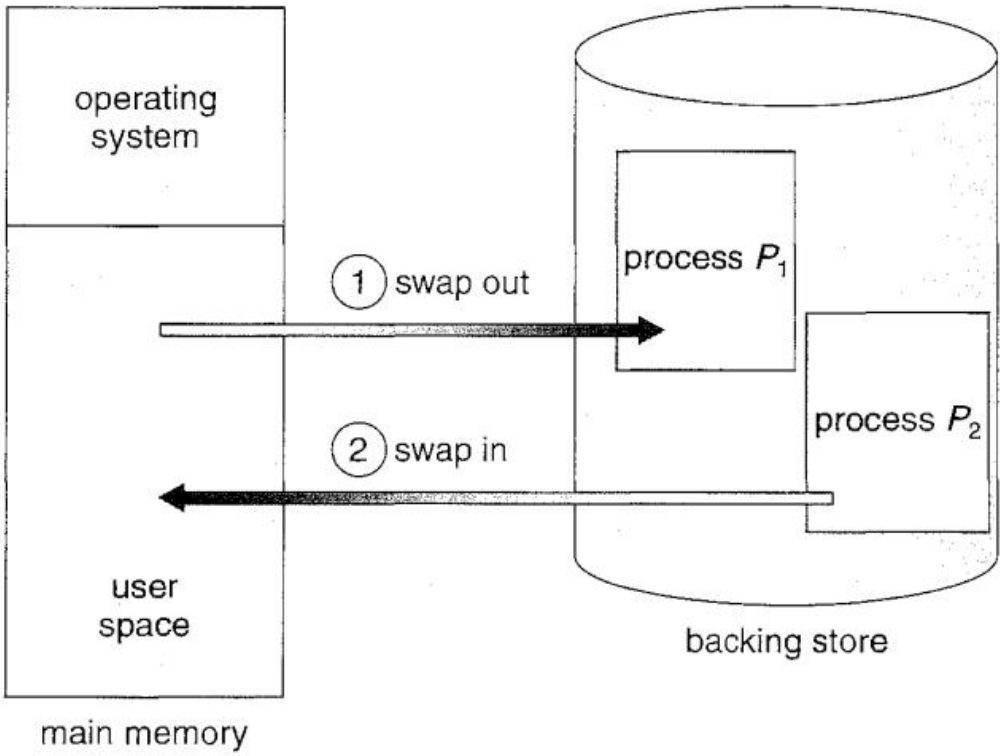

Figure 8.5 Swapping of two processes using a disk as a backing store.

lower-priority process can be swapped back in and continued. This variant of swapping is sometimes called roll out, roll in.

Normally, a process that is swapped out will be swapped back into the same memory space it occupied previously. This restriction is dictated by the method of address binding. If binding is done at assembly or load time, then the process cannot be easily moved to a different location. If execution-time binding is being used, however, then a process can be swapped into a different memory space, because the physical addresses are computed during execution time.

Swapping requires a backing store. The backing store is commonly a fast disk. It must be large enough to accommodate copies of all memory images for all users, and it must provide direct access to these memory images. The system maintains a ready queue consisting of all processes whose memory images are on the backing store or in memory and are ready to run. Whenever the CPU scheduler decides to execute a process, it calls the dispatcher. The dispatcher checks to see whether the next process in the queue is in memory. If it is not, and if there is no free memory region, the dispatcher swaps out a process currently in memory and swaps in the desired process. It then reloads registers and transfers control to the selected process. The context-switch time in such a swapping system is fairly high. To get an idea of the context-switch time, let us assume that the user process is 100 MB in size and the backing store is a standard hard disk with a transfer rate of 50MB per second. The actual transfer of the 100-MB process to or from main memory takes

100MB/50MB per second= 2 seconds.

Assuming an average latency of 8 milliseconds, the swap time is 2008 milliseconds. Since we must both swap out and swap in, the total swap time is about 4016 milliseconds. Notice that the major part of the swap time is transfer time. The total transfer time is directly proportional to the amount of memory swapped. If we have a computer system with 4 GB of main memory and a resident operating system taking 1 GB, the maximum size of the user process is 3GB. However, many user processes may be much smaller than this-say, 100 MB. A 100-MB process could be swapped out in 2 seconds, compared with the 60 seconds required for swapping 3 GB. Clearly, it would be useful to know exactly how much memory a user process is using, not simply how much it might be using. Then we would need to swap only what is actually used, reducing swap time. For this method to be effective, the user must keep the system informed of any changes in memory requirements. Thus, a process with dynamic memory requirements will need to issue system calls (request memory and release memory) to inform the operating system of its changing memory needs. Swapping is constrained by other factors as well. If we want to swap a process, we must be sure that it is completely idle. Of particular concern is any pending I/0. A process may be waiting for an I/0 operation when we want to swap that process to free up memory. However, if the I/0 is asynchronously accessing the user memory for I/0 buffers, then the process cannot be swapped. Assume that the I/0 operation is queued because the device is busy. If we were to swap out process P1 and swap in process P2, the I/0 operation might then attempt to use memory that now belongs to process P2 . There are two main solutions to this problem: never swap a process with pending I/0, or execute I/0 operations only into operating-system buffers. Transfers between operating-system buffers and process memory then occur only when the process is swapped in.

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4. Contiguous Memory Allocation

The main memory must accommodate both the operating system and the various user processes. We therefore need to allocate main menlOry in the most efficient way possible. This

section explains one common method, contiguous memory allocation. The memory is usually divided into two partitions: one for the resident operating system and one for the user processes. We can place the operating system in either low memory or high memory. The major factor affecting this decision is the location of the interrupt vector. Since the interrupt vector is often in low memory, programmers usually place the operating system in low memory as well. Thus, in this text, we discuss only the situation in which the operating system resides in low memory. The development of the other situation is similar. We usually want several user processes to reside in memory at the same time. We therefore need to consider how to allocate available memory to the processes that are in the input queue waiting to be brought into memory.

Memory Allocation:-

Now we are ready to turn to memory allocation. One of the simplest methods for allocating memory is to divide memory into several fixed-sized partition Each partition may contain exactly one process. Thus, the degree

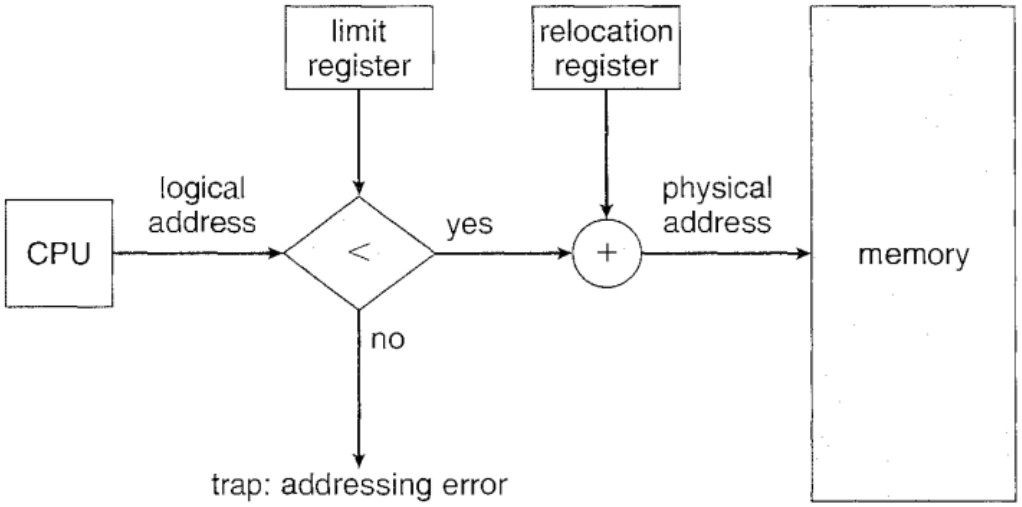

Figure 8.6 Hardware support for relocation and limit registers.

of multiprogramming is bound by the number of partitions. In this multi partition methodwhen a partition is free, a process is selected from the input queue and is loaded into the free partition. When the process terminates, the partition becomes available for another process. This method was originally used by the IBM OS/360 operating system (called MFT); it is no longer in use. The method described next is a generalization of the fixed-partition scheme (called MVT); it is used primarily in batch environments. Many of the ideas presented here are also applicable to a time-sharing environment in which pure segmentation is used for memory management (Section 8.6).

In the variable partitioning scheme, the operating system keeps a table indicating which parts of memory are available and which are occupied. Initially, all memory is available for user processes and is considered one large block of available memory a hole. Eventually as you will

see, memory contains a set of holes of various sizes. As processes enter the system, they are put into an input queue. The operating system takes into account the memory requirements of each process and the amount of available memory space in determining which processes are allocated memory. When a process is allocated space, it is loaded into memory, and it can then compete for CPU time. When a process terminates, it releases its memory which the operating system may then fill with another process from the input queue. At any given time, then, we have a list of available block sizes and an input queue. The operating system can order the input queue according to a scheduling algorithm. Memory is allocated to processes untit finally, the memory requirements of the next process cannot be satisfied -that is, no available block of memory (or hole) is large enough to hold that process. The operating system can then wait until a large enough block is available, or it can skip down the input queue to see whether the smaller memory requirements of some other process can be met. In generat as mentioned, the memory blocks available comprise a set of holes of various sizes scattered throughout memory. When a process arrives and needs memory, the system searches the set for a hole that is large enough for this process. If the hole is too large, it is split into two parts. One part is allocated to the arriving process; the other is returned to the set of holes. When a process terminates, it releases its block of memory, which is then placed back in the set of holes. If the new hole is adjacent to other holes, these adjacent holes are merged to form one larger hole. At this point, the system may need to check whether there are processes waiting for memory and whether this newly freed and recombined memory could satisfy the demands of any of these waiting processes.

- **First fit.** Allocate the first hole that is big enough. Searching can start either at the beginning of the set of holes or at the location where the previous first-fit search ended. We can stop searching as soon as we find a free hole that is large enough.
- **Best fit.** Allocate the smallest hole that is big enough. We must search the entire list, unless the list is ordered by size. This strategy produces the smallest leftover hole.
- **Worst fit**. Allocate the largest hole. Again, we must search the entire list, unless it is sorted by size. This strategy produces the largest leftover hole, which may be more useful than the smaller leftover hole from a best-fit approach.

Allocates a single contiguous block of memory to a process. Two main types:

- Fixed partitioning: Memory divided into fixed-size partitions; wastes memory due to internal fragmentation.
- Dynamic partitioning: Partitions created dynamically but may cause external fragmentation.

---

4. Segmentation

Memory divided into segments corresponding to logical divisions like code, stack, data. Each segment has a base and limit register.

- Allows better modularity and protection.
- Can cause external fragmentation.

Diagram: Segmentation

text

Process Segments:

Code Segment | Data Segment | Stack Segment

Mapped to Non-contiguous Memory Locations

Segment Table holds base and limit for each segment.

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5. Paging

Paging divides physical memory into fixed-size frames and logical memory into pages of the same size.

Paging is a memory-management scheme that permits the physical address space a process to be noncontiguous. Paging avoids external fragmentation and the need for compaction. It also solves the considerable problem of fitting memory chunks of varying sizes onto the backing store; most memory management schemes used before the introduction of paging suffered from this problem. The problem arises because, when some code fragments or data residing in main memory need to be swapped out, space must be fmmd on the backing store. The backing store has the same fragmentation problems discussed in connection with main memory, but access is much slower, so compaction is impossible. Because of its advantages over earlier methods, paging in its various forms is used in most operating systems.

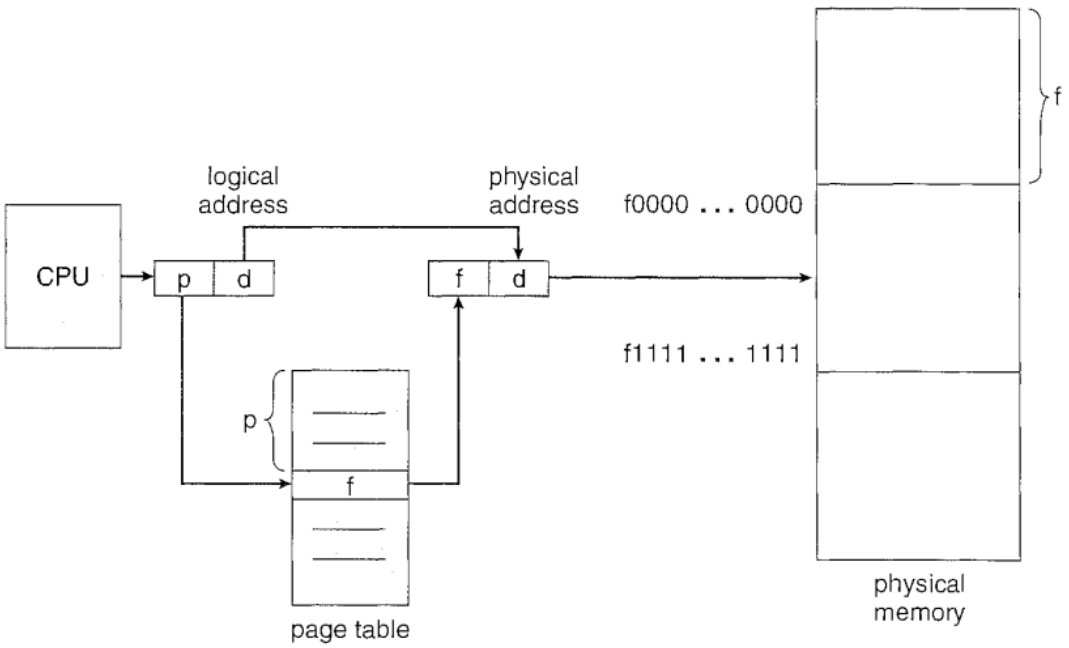

Figure 8.7 Paging hardware.

Traditionally, support for paging has been handled by hardware. However, recent designs have implemented paging by closely integrating the hardware and operating system, especially on 64-bit microprocessors.

8.4.1 Basic Method The basic method for implementing paging involves breaking physical memory into fixed-sized blocks called harnes and breaking logical memory into blocks of the same size called paging. When a process is to be executed, its pages are loaded into any available memory frames from their source (a file system or the backing store). The backing store is divided into fixed-sized blocks that are of the san1.e size as the memory frames.

The hardware support for paging is illustrated in Figure 8.7. Every address generated by the CPU is divided into two parts: a page number(p) and a page offset (d) The page number is used as an index into a page table.

The page table contains the base address of each page in physical memory. This base address is combined with the page offset to define the physical memory address that is sent to the memory unit. The paging model of memory is shown in Figure 8.8.

The page size (like the frame size) is defined by the hardware. The size of a page is typically a power of 2, varying between 512 bytes and 16 MB per page, depending on the computer architecture. The selection of a power of 2 as a page size makes the translation of a logical address into a page number and page offset particularly easy. If the size of the logical address space is 2m, and a page size is 271 addressing units (bytes or wordst then the high-order m- n bits of a logical address designate the page number, and the n low-order bits designate the page offset. Thus, the logical address is as follows:

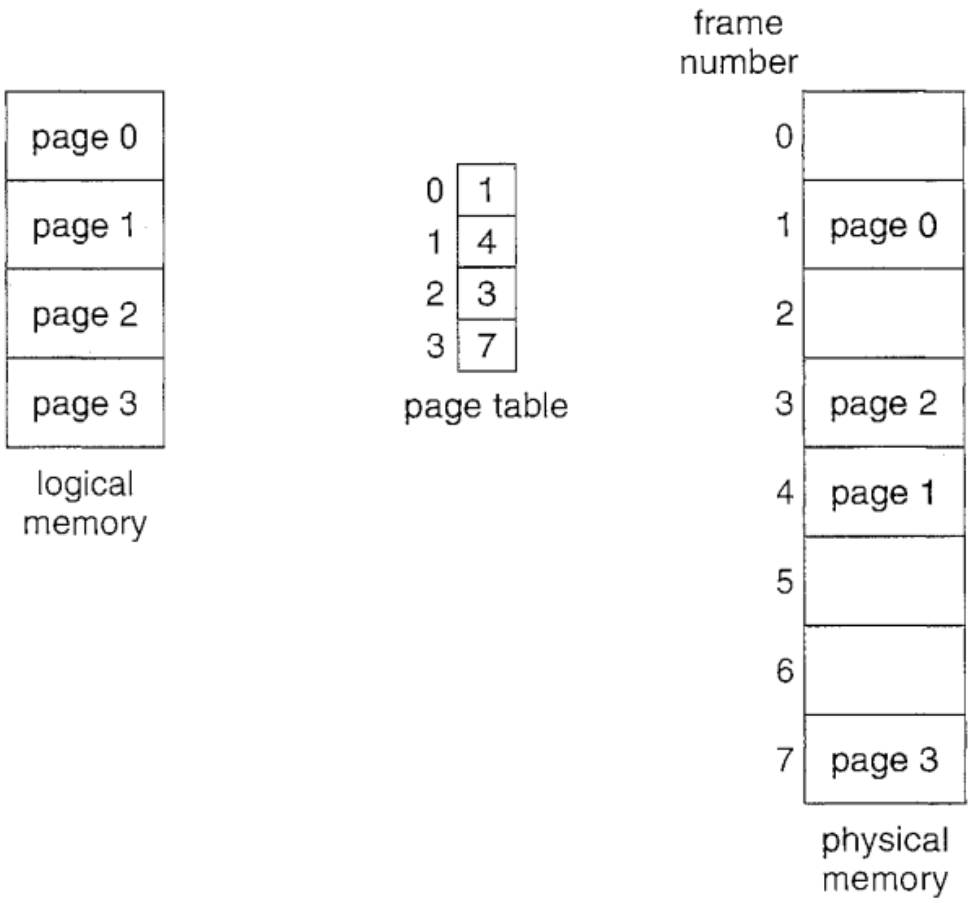

Figure 8.8 Paging model of logical and physical memory.

page number	page offset
p	d
$m - n$	n

where p is an index into the page table and d is the displacement within the page. As a concrete (although minuscule) example, consider the memory in Figure 8.9. Here, in the logical address, n= 2 and m = 4. Using a page size of 4 bytes and a physical memory of 32 bytes (8 pages), we show how the user's view of memory can be mapped into physical memory. Logical address 0 is page 0, offset 0. Indexing into the page table, we find that page 0 is in frame 5. Thus, logical address 0 maps to physical address 20 [= (5 x 4) + 0]. Logical address 3 (page 0, offset 3) maps to physical address 23 [= (5 x 4) + 3]. Logical address 4 is page 1, offset 0; according to the page table, page 1 is mapped to frame 6. Thus, logical address 4 maps to physical address 24 [= (6 x 4) + O]. Logical address 13 maps to physical address 9. You may have noticed that paging itself is a form of dynamic relocation. Every logical address is bound by the paging hardware to some physical address. Using paging is similar to using a table of base (or relocation) registers, one for each frame of memory.

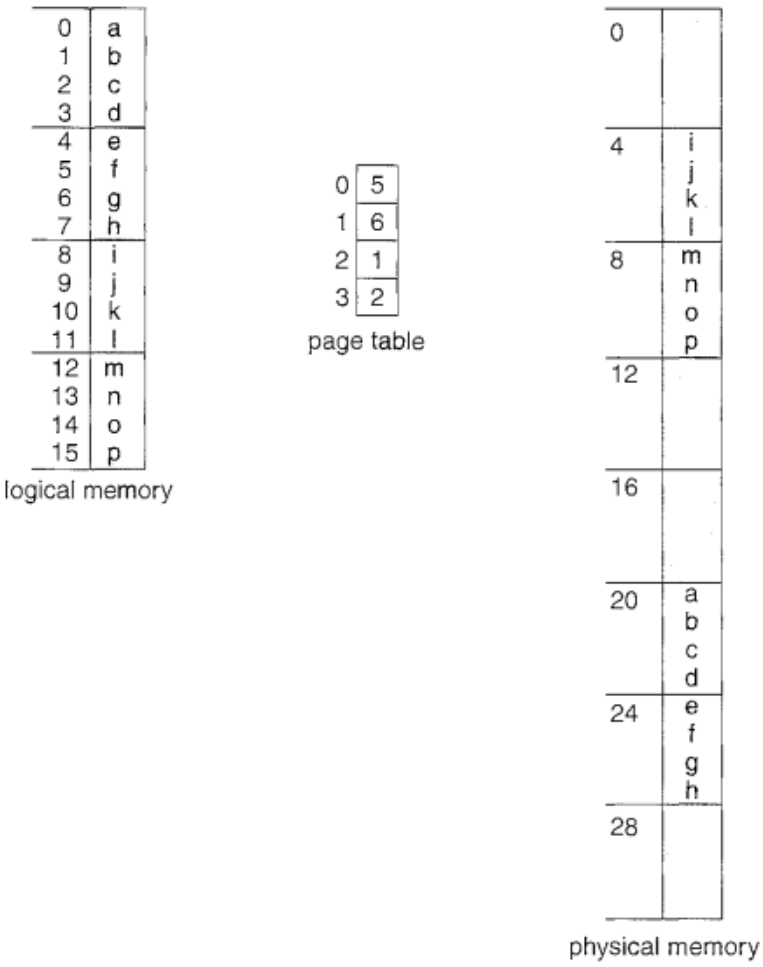

Figure 8.9 Paging example for a 32-byte memory with 4-byte pages.

When we use a paging scheme, we have no external fragmentation: any free frame can be allocated to a process that needs it. However, we may have some internal fragmentation. Notice that frames are allocated as units. If the memory requirements of a process do not happen to coincide with page boundaries, the last frame allocated may not be completely full. For example, if page size is 2,048 bytes, a process of 72,766 bytes will need 35 pages plus 1,086 bytes. It will be allocated 36 frames, resulting in internal fragmentation of 2,048 - 1,086 = 962 bytes. In the worst case, a process would need 11 pages plus 1 byte. It would be allocated 11 + 1 frames, resulting in internal fragmentation of almost an entire frame.

If process size is independent of page size, we expect internal fragmentation to average one-half page per process. This consideration suggests that small page sizes are desirable. However, overhead is involved in each page-table entry, and this overhead is reduced as the size of the pages increases. Also, disk I/0 is more efficient when the amount data being transferred is larger (Chapter 12). Generally, page sizes have grown over time as processes, data sets, and main memory have become larger. Today, pages typically are between 4 KB and 8 KB in size, and some systems support even larger page sizes. Some CPUs and kernels even support multiple page sizes. For instance, Solaris uses page sizes of 8 KB and 4 MB, depending on the data stored by the pages. Researchers are now developing support for variable on-the-fly page size.

Usually, each page-table entry is 4 bytes long, but that size can vary as well. A 32-bit entry can point to one of 232 physical page frames. If frame size is 4 KB, then a system with 4-byte entries can address 244 bytes (or 16 TB) of physical memory.

When a process arrives in the system to be executed, its size, expressed in pages, is examined. Each page of the process needs one frame. Thus, if the process requires 11 pages, at least 11 frames must be available in memory. If n frames are available, they are allocated to this arriving process. The first page of the process is loaded inJo one of the allocated frames, and the frame number is put in the page table for this process. The next page is loaded into another frame, its frame number is put into the page table, and so on (Figure 8.10).

An important aspect of paging is the clear separation between the user's view of memory and the actual physical memory. The user program views memory as one single space, containing only this one program. In fact, the user program is scattered throughout physical memory, which also holds other programs. The difference between the user's view of memory and the actual physical memory is reconciled by the address-translation hardware. The logical addresses are translated into physical addresses. This mapping is hidden from the user and is controlled by the operating system. Notice that the user process by definition is unable to access memory it does not own. It has no way of addressing memory outside of its page table, and the table includes only those pages that the process owns. Since the operating system is managing physical memory, it must be aware of the allocation details of physical memory-which frames are allocated, which frames are available, how many total frames there are, and so on. This information is generally kept in a data structure called a frame table.

8.4.2 Hardware Support Each operating system has its own methods for storing page tables. Most allocate a page table for each process. A pointer to the page table is stored with the other register values (like the instruction counter) in the process control block. When the dispatcher is told to start a process, it must reload the user registers and define the correct hardware page-table values from the stored user page table.

The hardware implementation of the page table can be done in several ways. In the simplest case, the page table is implemented as a set of dedicated registers. These registers should be built with very high-speed logic to make the paging-address translation efficient. Every access to memory nlust go through the paging map, so efficiency is a major consideration. The CPU dispatcher reloads these registers, just as it reloads the other registers. Instructions to load or modify the page-table registers are, of course, privileged, so that only the operating system can change the memory map. The DEC PDP-11 is an example of such an architecture. The address consists of 16 bits, and the page size is 8 KB. The page table thus consists of eight entries that are kept in fast registers.

The use of registers for the page table is satisfactory if the page table is reasonably sncall (for example, 256 entries). Most contemporary computers, however, allow the page table to be very large (for example, 1 million entries). For these machines, the use of fast registers to implement the page table is not feasible. Rather, the page table is kept in main memory, and a page table base register(PTBR) points to the page table. Changing page tables requires changing only this one register, substantially reducing context-switch time. The problem with this approach is the time required to access a user memory location. If we want to access location i, we must first index into the page table, using the value in the PTBR offset by the page number fori. This task requires a memory access. It provides us with the frame number, which is combined with the page offset to produce the actual address. We can then access the desired place in memory. With this scheme, two memory accesses are needed to access a byte (one for the page-table

entry, one for the byte). Thus, memory access is slowed by a factor of 2. This delay would be intolerable under most circumstances. We might as well resort to swapping!

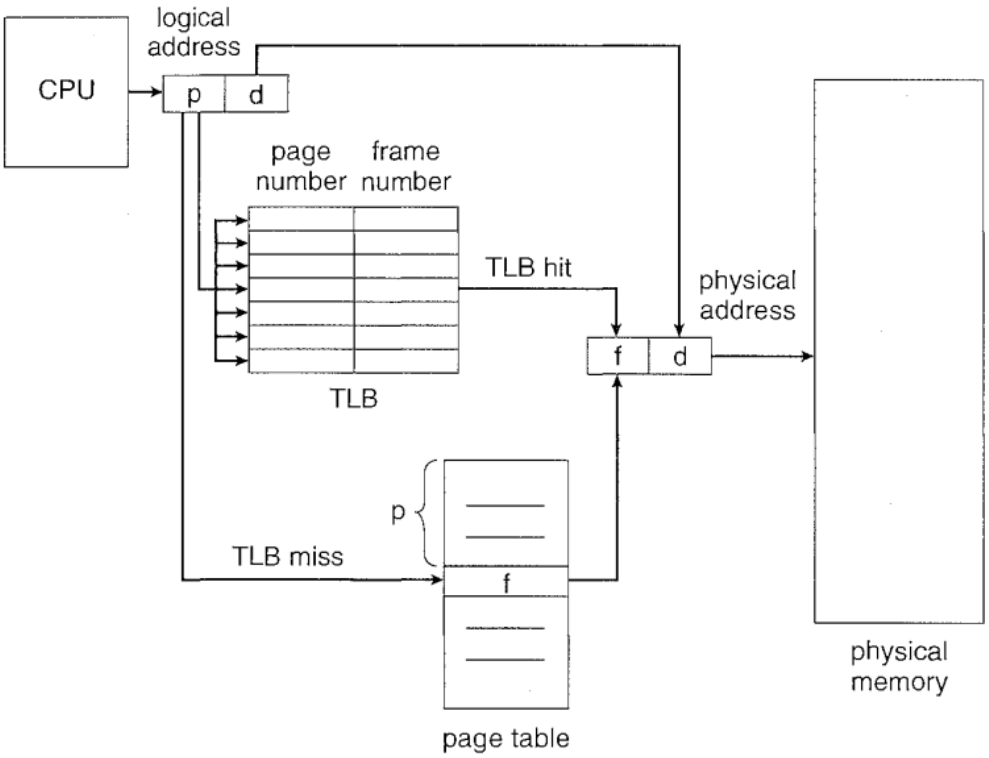

Figure 8.11 Paging hardware with TLB.

contain valid virtual addresses but have incorrect or invalid physical addresses left over from the previous process. The percentage of times that a particular page number is found in the TLB is called the hit ratio. An 80-percent hit ratio, for example, means that we find the desired page number in the TLB 80 percent of the time. If it takes 20 nanoseconds to search the TLB and 100 nanoseconds to access memory, then a mapped-memory access takes 120 nanoseconds when the page number is in the TLB. If we fail to find the page number in the TLB (20 nanoseconds), then we must first access memory for the page table and frame number (100 nanoseconds) and then access the desired byte in memory (100 nanoseconds), for a total of 220 nanoseconds. To find the effective memory access time

we weight the case by its probability: effective access time = 0.80 x 120 + 0.20 x 220 = 140 nanoseconds. In this example, we suffer a 40-percent slowdown in memory-access time (from 100 to 140 nanoseconds). For a 98-percent hit ratio, we have effective access time = 0.98 x 120 + 0.02 x 220 = 122 nanoseconds. This increased hit rate produces only a 22 percent slowdown in access time. We will further explore the impact of the hit ratio on the TLB in Chapter 9. 8.

8.4.3 Protection

Memory protection in a paged environment is accomplished by protection bits associated with each frame. Normally, these bits are kept in the page table. One bit can define a page to be read-write or read-only. Every reference to memory goes through the page table to find the correct frame nuncber. At the same time that the physical address is being computed, the protection bits can be checked to verify that no writes are being made to a read-only page. An attempt to write to a read-only page causes a hardware trap to the operating system (or memory-

protection violation). We can easily expand this approach to provide a finer level of protection. We can create hardware to provide read-only, read-write, or execute-only protection; or, by providing separate protection bits for each kind of access, we can allow any combination of these accesses. Illegal attempts will be trapped to the operating system. One additional bit is generally attached to each entry in the page table: a valid – invalid bit. When this bit is set to "valid," the associated page is in the process's logical address space and is thus a legal (or valid) page. When the bit is set to"invalid," the page is not in the process's logical address space. Illegal addresses are trapped by use of the valid -invalid bit. The operating system sets this bit for each page to allow or disallow access to the page. Suppose, for example, that in a system with a 14-bit address space (0 to 16383), we have a program that should use only addresses 0 to 10468. Given a page size of 2 KB, we have the situation shown in Figure 8.12. Addresses in

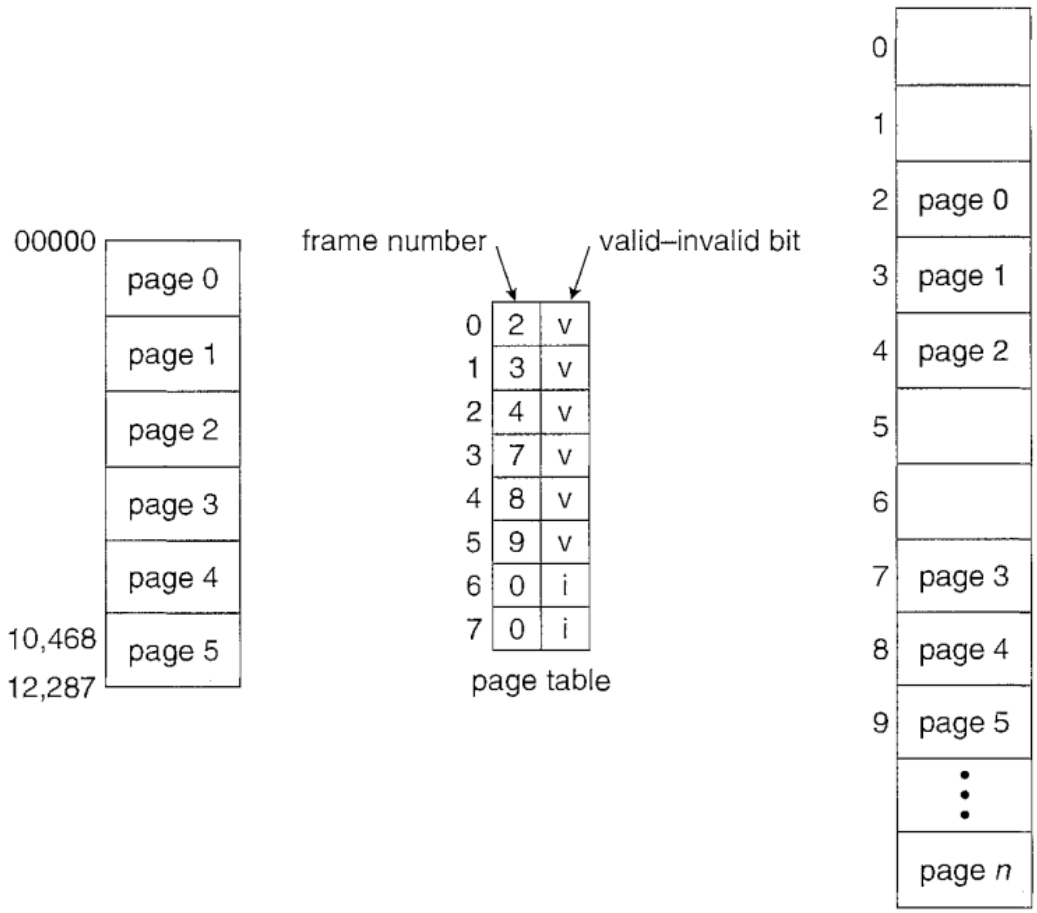

Figure 8.12 Valid (v) or invalid (i) bit in a page table.

pages 0, 1, 2, 3, 4, and 5 are mapped normally through the page table. Any attempt to generate an address in pages 6 or 7, however, will find that the valid -invalid bit is set to invalid, and the computer will trap to flee operating system (invalid page reference). Notice that this scheme has created a problem. Because the program extends only to address 10468, any reference beyond that address is illegal. Howeve1~ references to page 5 are classified as valid, so accesses to addresses up to 12287 are valid. Only the addresses from 12288 to 16383 are invalid. This problem is a result of the 2-KB page size and reflects the internal fragmentation of paging. Rarely does a process use all its address range. In fact many processes use only a

small fraction of the address space available to them. It would be wasteful in these cases to create a page table with entries for every page in the address range. Most of this table would be unused but would take up valuable memory space. Some systems provide hardware, in the form of a page table in length register(PTLR). to indicate the size of the page table. value is checked against every logical address to verify that the address is in the valid range for the process. Failure of this test causes an error trap to the operating system.

8.4.4 Shared Pages

An advantage of paging is the possibility of sharing common code. This consideration is particularly important in a time-sharing environment. Consider a system that supports 40 users, each of whom executes a text editor. If the text editor consists of 150 KB of code and 50 KB of data space, we need 8,000 KB to support the 40 users. If the code is (or pure however, it can be shared, as shown in Figure 8.13. Here we see a three-page editor-each page 50 KB in size (the large page size is used to simplify the figure)-being shared among three processes. Each process has its own data page. Reentrant code is non-self-modifying code: it never changes during execution. Thus, two or more processes can execute the same code at the same time. Each process has its own copy of registers and data storage to hold the data for the process's execution. The data for two different processes wilt of course, be different. Only one copy of the editor need be kept in physical memory. Each user's page table maps onto the same physical copy of the editor, but data pages are mapped onto different frames. Thus, to support 40 users, we need only one copy of the editor (150 KB), plus 40 copies of the 50 KB of data space per user. The total space required is now 2)50 KB instead of 8,000 KB-a significant savings. Other heavily used programs can also be shared -compilers, window systems, run-time libraries, database systems, and so on. To be sharable, the code must be reentrant. The read-only nature of shared code should not be left to the correctness of the code; the operating system should enforce this property. The sharing of memory among processes on a system is similar to the sharing of the address space of a task by threads, described in Chapter 4. Furthermore, recall that in Chapter 3 we described shared memory as a method

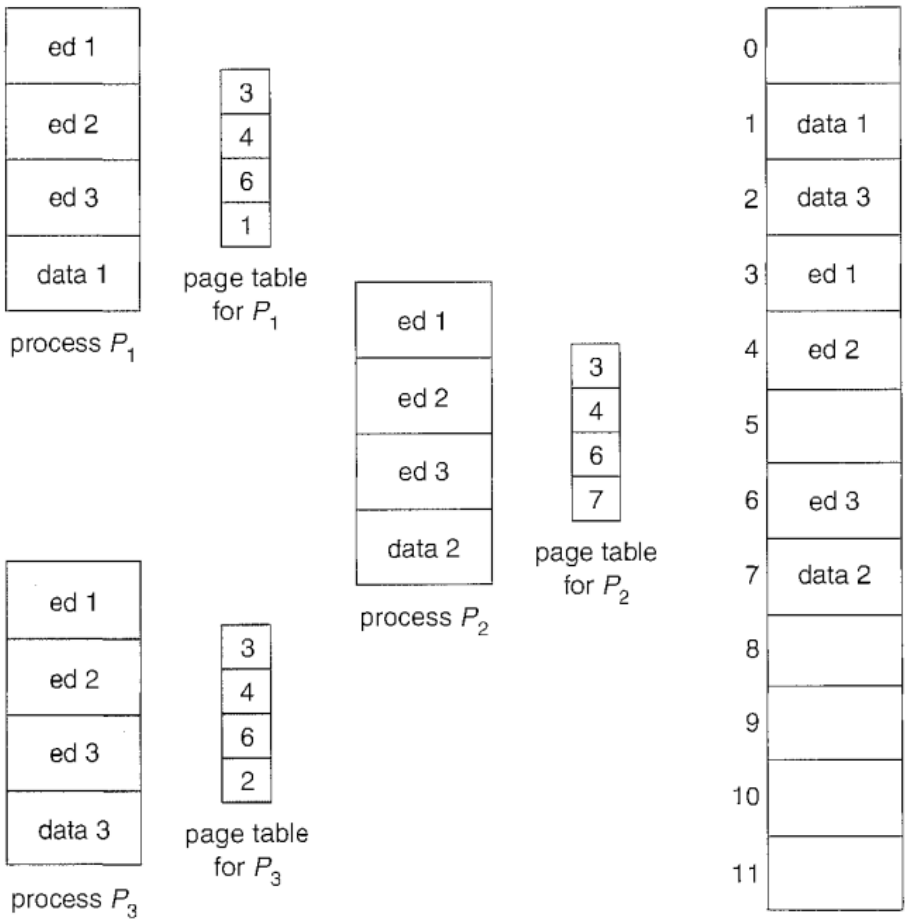

Figure 8.13 Sharing of code in a paging environment.

- Eliminates external fragmentation.
- Uses page tables to map logical pages to physical frames.
- Allows non-contiguous memory allocation.

5. Segmentation:-

An. important aspect of memory management that became unavoidable with paging is the separation of the user's view of memory from the actual physical memory. As we have already seen, the user's view of memory is not the same as the actual physical memory. The user's view is mapped onto physical memory. This mapping allows differentiation between logical memory and physical memory.

8.6.1 Basic Method Do users think of memory as a linear array of bytes, some containing instructions and others containing data? Most people would say no. Rather, users prefer to view memory as a collection of variable-sized segments, with no necessary ordering among segments (Figure 8.18). Consider how you think of a program when you are writing it. You think of it as a main program with a set of methods, procedures, or functions. It may also include various data structures: objects, arrays, stacks, variables, and so on. Each of these modules or data elements is referred to by name. You talk about "the stack," "the math library," "the n1.ain program," without caring what addresses in memory these elements occupy. You are not concerned with whether the stack is stored

before or after the Sqrt () function. Each of these segments is of variable length; the length is intrinsically defined by

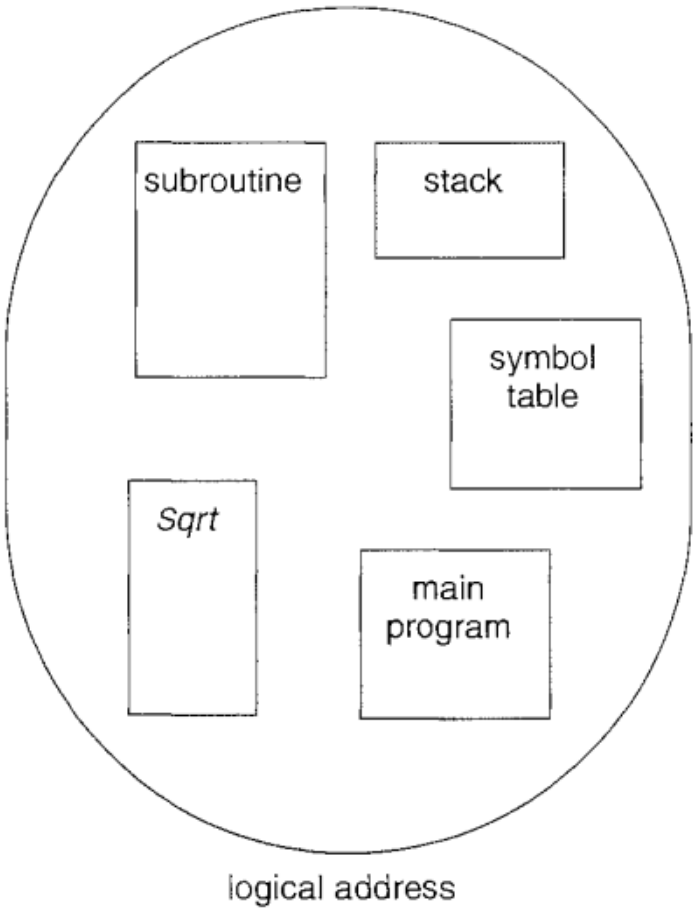

Figure 8.18 User's view of a program.

the purpose of the segment in the program. Elements within a segment are identified by their offset from the begim1.ing of the segment: the first statement of the program, the seventh stack frame entry in the stack, the fifth instruction of the Sqrt (), and so on. is a memory-management scheme that supports this user view of memory. A logical address space is a collection of segments. Each segment has a name and a length. The addresses specify both the segment name and the offset within the segment. The user therefore specifies each address by two quantities: a segment name and an offset. (Contrast this scheme with the paging scheme, in which the user specifies only a single address, which is partitioned by the hardware into a page number and an offset, all invisible to the programmer.) For simplicity of implementation, segments are numbered and are referred to by a segn"lent number, rather than by a segment name. Thus, a logical address consists of a two tuple: . Normally, the user program is compiled, and the compiler automatically constructs segments reflecting the input program.

A C compiler might create separate segments for the following:

- The code
- Global variables
- The heap, from which memory is allocated
- The stacks used by each thread
- The standard C library

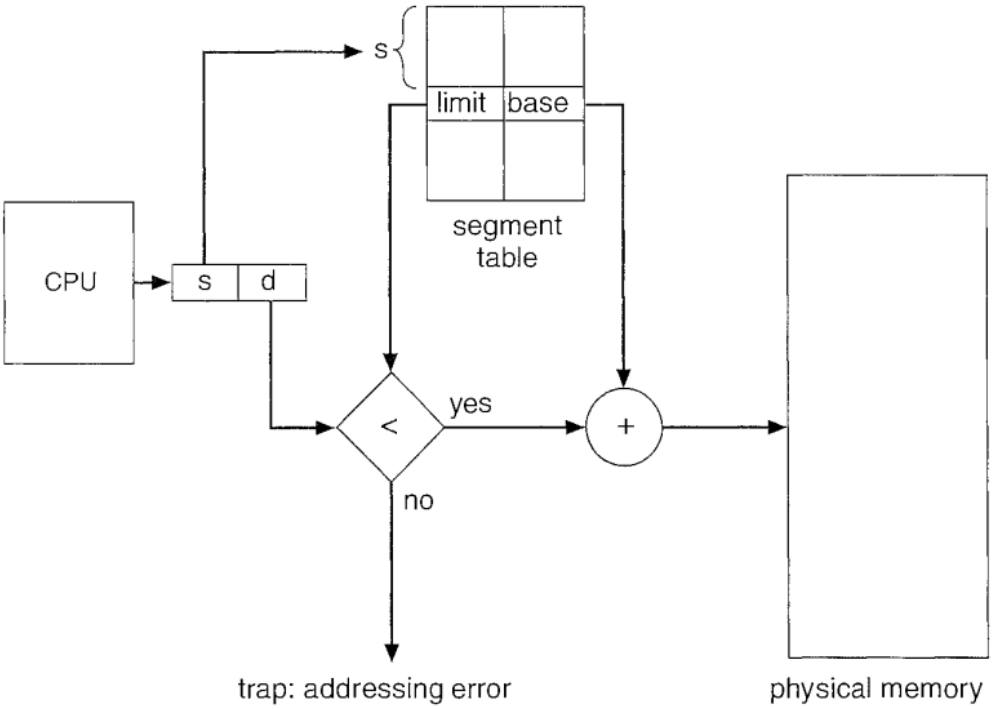

Figure 8.19 Segmentation hardware.

Libraries that are linked in during compile time might be assign.ed separate segments. The loader would take all these segments and assign them segment numbers.

8.6.2 Hardware

Although the user can now refer to objects in the program by a two-dimensional address, the actual physical memory is still, of course, a one-dimensional sequence of bytes. Thus, we must define an implementation to map twodimensional user-defined addresses into one-dimensional physical addresses. This mapping is effected by a Each entry in the segment table has a segment base and a segment limit. The segment base contains the startilcg physical address where the segment resides in memory, and the segment limit specifies the length of the segment. The use of a segment table is illustrated in Figure 8.19. A logical address consists of two parts: a segment number, s, and an offset into that segment, d. The segment number is used as an index to the segment table. The offset d of the logical address must be between 0 and the segment limit. If it is not, we trap to the operating system (logical addressing attempt beyond end of segment). When an offset is legal, it is added to the segment base to produce the address in physical memory of the desired byte. The segment table is thus essentially an array of base-limit register pairs. As an example, consider the situation shown in Figure 8.20. We have five segments numbered from 0 through 4. The segments are stored in physical memory as shown. The segment table has a separate entry for each segment, giving the beginning address of the segment in physical memory (or base) and the length of that segment (or limit). For example, segment 2 is 400 bytes long and begins at location 4300. Thus, a reference to byte 53 of segment 2 is mapped

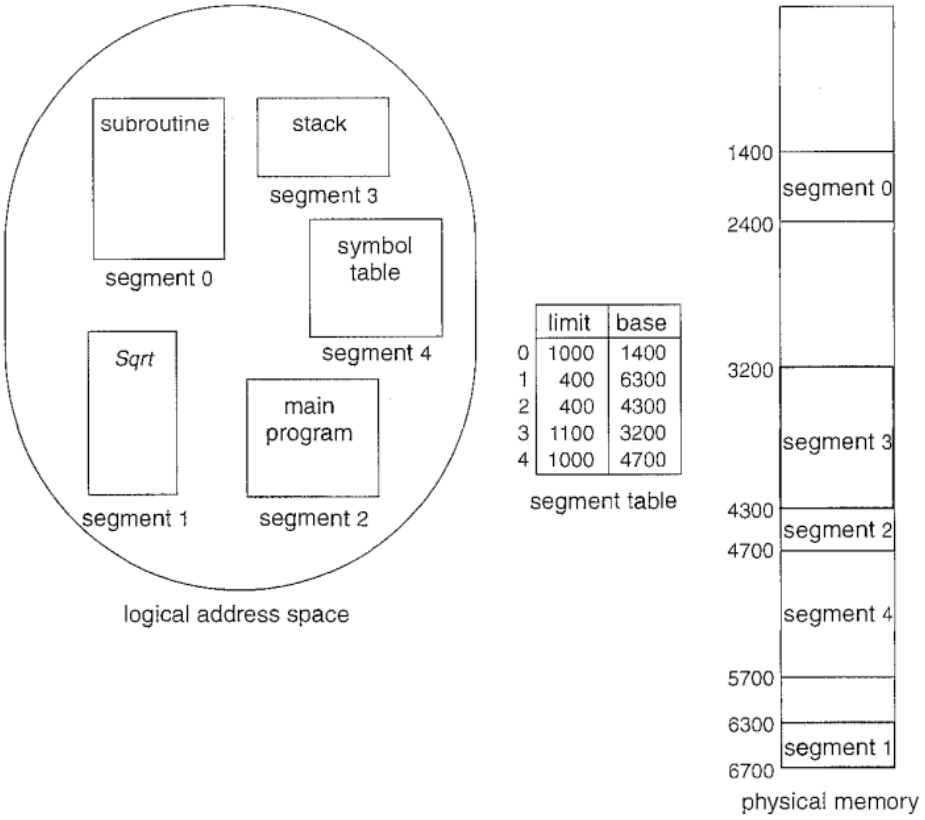

Figure 8.20 Example of segmentation.

onto location 4300 +53= 4353. A reference to segment 3, byte 852, is mapped to 3200 (the base of segment 3) + 852 = 4052. A reference to byte 1222 of segment 0 would result in a trap to the operating system, as this segment is only tOOO bytes long.

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6. Virtual Memory Management

- Allows running programs larger than physical memory by loading pages on demand.
- **<u>Uses demand paging:</u>** pages loaded when referenced.

  Consider how an executable program might be loaded from disk into n'lemory. One option is to load the entire program in physical memory at program execution time. However, a problent with this approach is that we may not initially need the entire program in memory. Suppose a program starts with a list of available options from which the user is to select. Loading the entire program into memory results in loading the executable code for all options, regardless of whether an option is ultimately selected by the user or not. An alternative strategy is to load pages only as they are needed. This technique is known as demand page and is commonly used in virtual memory systems. With demand-paged virtual memory, pages are only loaded when they are demanded during program execution; pages that are never accessed are thus never loaded into physical memory. A demand-paging system is similar to a paging system with swapping (Figure 9.4) where processes reside in secondary memory (usually a

disk). When we want to execute a process, we swap it into memory. Rather than swapping the entire process into memory, however, we use a lazy swapper A lazy swapper never swaps a page into memory unless that page will be needed. Since we are now viewing a process as a sequence of pages, rather than as one large contiguous address space, use of the term swapper is technically incorrect. A swapper manipulates entire processes, whereas a pager is concerned with the individual pages of a process. We thus use pager, rather than swapper, in connection with demand paging.

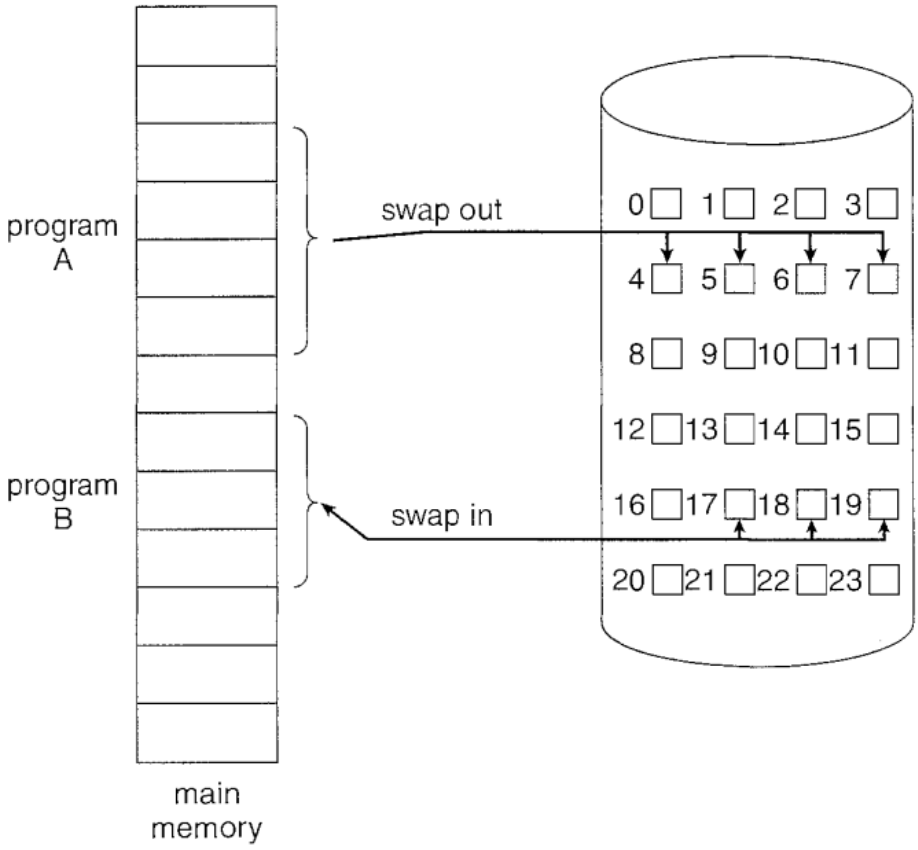

Figure 9.4 Transfer of a paged memory to contiguous disk space.

9.2.1 Basic Concepts When a process is to be swapped in, the pager guesses which pages will be used before the process is swapped out again. Instead of swapping in a whole process, the pager brings only those pages into memory. Thus, it avoids reading into memory pages that will not be used anyway, decreasing the swap time and the amount of physical memory needed. With this scheme, we need some form of hardware support to distinguish between the pages that are in memory and the pages that are on the disk. The valid -invalid bit scheme described in Section 8.4.3 can be used for this purpose. This time, however, when this bit is set to "valid/' the associated page is both legal and in n1.emory. If the bit is set to "invalid/' the page either is not valid (that is, not in the logical address space of the process) or is valid but is currently on the disk. The page-table entry for a page that is brought into memory is set as usuat but the page-table entry for a page that is not currently in memory is either simply marked invalid or contains the address of the page on disk. This situation is depicted in Figure 9.5. Notice that marking a page invalid will have no effect if the process never attempts to access

that page. Hence, if we guess right and page in all and only those pages that are actually needed, the process will run exactly as though we had brought in all pages. While the process executes and accesses pages that are memory resident, execution proceeds normally.

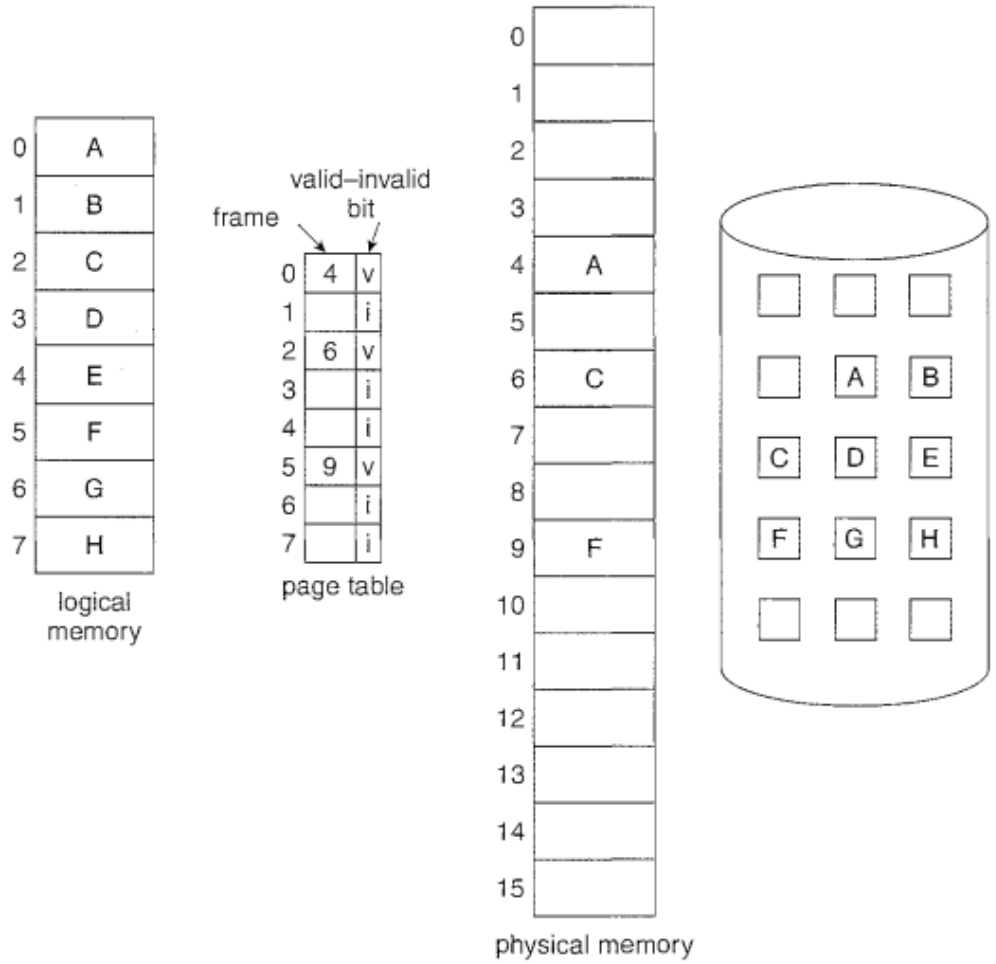

Figure 9.5 Page table when some pages are not in main memory.

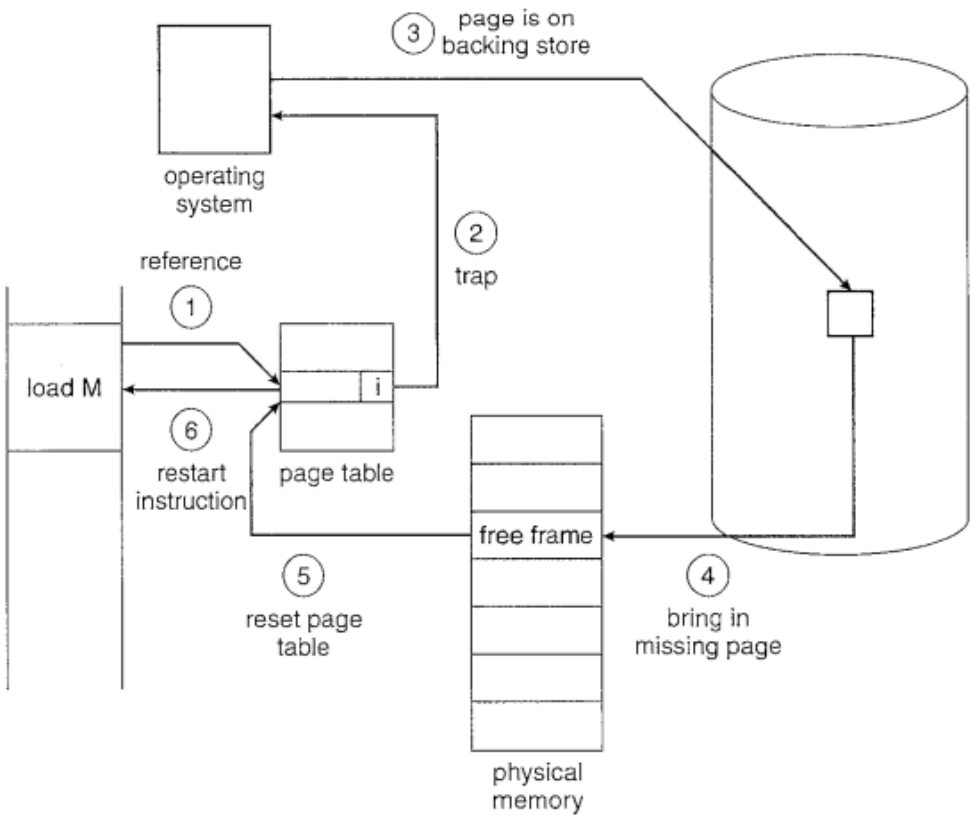

Figure 9.6 Steps in handling a page fault.

But what happens if the process tries to access a page that was not brought into memory? Access to a page marked invalid causes a page fault The paging hardware, in translating the address through the page table, will notice that the invalid bit is set, causing a trap to the operating system. This trap is the result of the operating system's failure to bring the desired page into memory. The procedure for handling this page fault is straightforward (Figure 9.6):

- We check an internal table (usually kept with the process control block) for this process to determine whether the reference was a valid or an invalid memory access.
- If the reference was invalid, we terminate the process. If it was valid, but we have not yet brought in that page, we now page it in.
- We find a free frame (by taking one from the free-frame list, for example). We schedule a disk operation to read the desired page into the newly allocated frame.
- When the disk read is complete, we modify the internal table kept with the process and the page table to indicate that the page is now in memory.
- We restart the instruction that was interrupted by the trap.
- The process can now access the page as though it had always been in memory.

Theoretically, some programs could access several new pages of memory with each instruction execution (one page for the instruction and many for data), possibly causing multiple page faults per instruction. This situation would result in unacceptable system performance. Fortunately, analysis of running processes shows that this behavior is exceedingly unlikely. Programs tend to have locality of reference, described in Section 9.6.1, which results in reasonable performance from demand paging.

The hardware to support demand paging is the same as the hardware for paging and swapping:

- **Page table**. This table has the ability to mark an entry invalid through a valid–invalid bit or a special value of protection bits.
- **Secondary memory**. This memory holds those pages that are not present in main memory. The secondary memory is usually a high-speed disk. It is known as the swap device, and the section of disk used for this purpose is known as swap space. Swap-space allocation is discussed in Chapter 12.

A crucial requirement for demand paging is the ability to restart any instruction after a page fault. Because we save the state (registers, condition code, instruction counter) of the interrupted process when the page fault occurs, we must be able to restart the process in *exactly* the same place and state, except that the desired page is now in memory and is accessible. In most cases, this requirement is easy to meet. A page fault may occur at any memory reference. If the page fault occurs on the instruction fetch, we can restart by fetching the instruction again. If a page fault occurs while we are fetching an operand, we must fetch and decode the instruction again and then fetch the operand.

As a worst-case example, consider a three-address instruction such as ADD the content of A to B, placing the result in C. These are the steps to execute this instruction:

1. Fetch and decode the instruction (ADD).
2. Fetch A.
3. Fetch B.
4. Add A and B.
5. Store the sum in C.

If we fault when we try to store in C (because C is in a page not currently in memory), we will have to get the desired page, bring it in, correct the page table, and restart the instruction. The restart will require fetching the instruction again, decoding it again, fetching the two operands again, and

<u>copy-on-write:-</u> In Section 9 .2, we illustrated how a process can start quickly by merely demandpaging in the page containing the first instruction. However, process creation using the fork() system call may initially bypass the need for demand paging by using a technique similar to page sharing (covered in Section 8.4.4). This technique provides for rapid process creation and minimizes the number of new pages that must be allocated to the newly created process.

Recall thatthe fork() system call creates a child process that is a duplicate of its parent. Traditionally, fork() worked by creating a copy of the parent's address space for the child, duplicating the pages belonging to the parent. However, considering that many child processes invoke the exec() system call immediately after creation, the copying of the parent's address space may be unnecessary. Instead, we can use a technique known as copy-on-write which works by allowing the parent and child processes initially to share the same pages.

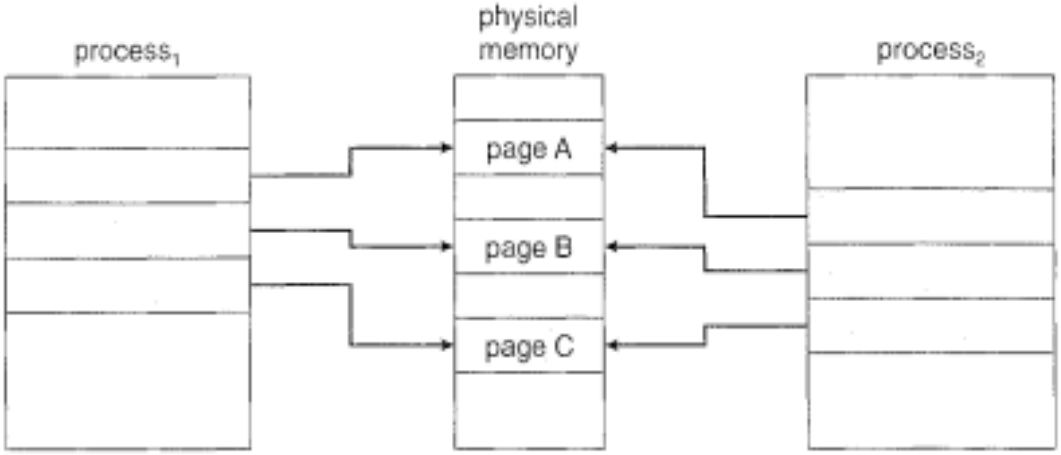

Figure 9.7 Before process 1 modifies page C.

These shared pages are marked as copy-on-write pages, meaning that if either process writes to a shared page, a copy of the shared page is created. Copy-on-write is illustrated in Figures 9.7 and Figure 9.8, which show the contents of the physical memory before and after process 1 modifies page c. For example, assume that the child process attempts to modify a page containing portions of the stack, with the pages set to be copy-on-write. The operating system will create a copy of this page, nl.apping it to the address space of the child process. The child process will then modify its copied page and not the page belonging to the parent process. Obviously, when the copy-on-write technique is used, only the pages that are modified by either process are copied; all unmodified pages can be shared by the parent and child processes. Note, too,

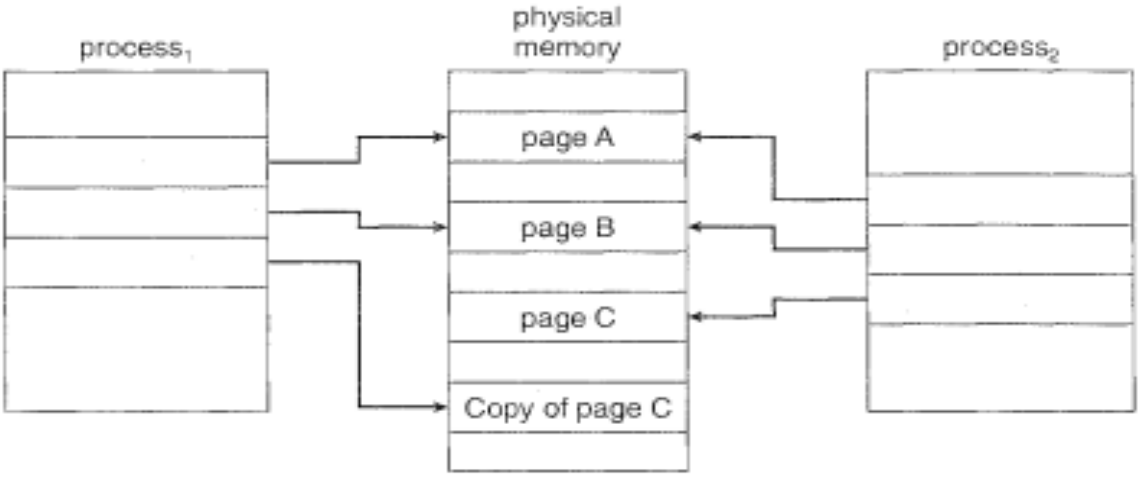

Figure 9.8 After process 1 modifies page C.

that only pages that can be nwdified need be m~arked as copy-on-write. Pages that cannot be modified (pages containing executable code) can be shared by the parent and child. Copy-on-write is a common technique used by several operating systems, including Windows XP, Linux, and Solaris. When it is determined that a page is going to be duplicated using copyon-write, it is important to note the location from which the free page will be allocated. Many operating systems provide a of free pages for such requests. These free pages are typically allocated when

the stack or heap for a process must expand or when there are copy-on-write pages to be managed. Operating systems typically allocate these pages using a technique known as zem-fHl-on-den:1and. Zero-fill-on-demand pages have been zeroed-out before being allocated, thus erasing the previous contents. Several versions of UNIX (including Solaris and Linux) provide a variation ofthe fork() system call-vfork() (for fori()- that operates differently from fork() with copy-on-write. With vfork(), the parent process is suspended, and the child process uses the address space of the parent. Because vfork() does not use copy-on-write, if the child process changes any pages of the parent's address space, the altered pages will be visible to the parent once it resumes. Therefore, vf ork () must be used with caution to ensure that the child process does not modify the address space of the parent. vf or k () is intended to be used when the child process calls exec() immediately after creation. Because no copying of pages takes place, vf ork () is an extremely efficient method of process creation and is sometimes used to implement UNIX command-line shell interfaces.

- Supports copy-on-write to efficiently manage shared pages.

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6. Page Replacement Algorithms

In our earlier discussion of the page-fault rate, we assumed that each page faults at most once, when it is first referenced. This representation is not strictly accurate, however. If a process of ten pages actually uses only half of them, then demand paging saves the I/0 necessary to load the five pages that are never used. We could also increase our degree of multiprogramming by running twice as many processes. Thus, if we had forty frames, we could run eight processes, rather than the four that could run if each required ten frames (five of which were never used). If we increase our degree of multiprogramming, we are memory. If we run six processes, each of which is ten pages in size but uses only five pages, we have higher CPU utilization and throughput, ten frames to spare. It is possible, however, that each of these processes, for a particular data set, may suddenly try to use all ten of its pages, resulting in a need for sixty frames when only forty are available. Further, consider that system memory is not used only for holding program pages. Buffers for I/ 0 also consume a considerable amount of memory. This use can increase the strain on memory-placement algorithms. Deciding how much memory to allocate to I/0 and how much to program pages is a significant challenge. Some systems allocate a fixed percentage of memory for I/0 buffers, whereas others allow both user processes and the I/0 subsystem to compete for all system memory.

When memory is full, a page must be replaced to load new ones. Algorithms:

- FIFO: First-in, first-out.

9.4.2 FIFO Page Replacement

The simplest page-replacement algorithm is a first-in, first-out (FIFO) algorithm. A FIFO replacement algorithm associates with each page the time when that page was brought into memory. When a page must be replaced, the oldest page is chosen. Notice that it is not strictly necessary to record the time when a page is brought in. We can create a FIFO queue to hold all pages in memory. We replace the page at the head of the queue. When a page is brought into memory, we insert it at the tail of the queue.

For our example reference string, our three frames are initially empty. The first three references (7, 0, 1) cause page faults and are brought into these empty frames. The next reference (2) replaces page 7, because page 7 was brought in first. Since 0 is the next reference and 0 is already in memory, we have no fault for this reference. The first reference to 3 results in replacement of page 0, since

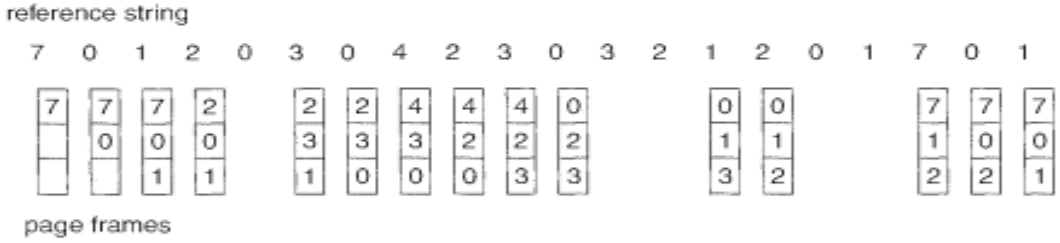

Figure 9.12 FIFO page-replacement algorithm.

it is now first in line. Because of this replacement, the next reference, to 0, will fault. Page 1 is then replaced by page 0. This process continues as shown in Figure 9.12. Every time a fault occurs, we show which pages are in our three frames. There are fifteen faults altogether.

The FIFO page-replacement algorithm is easy to understand and program. However, its performance is not always good. On the one hand, the page replaced may be an initialization module that was used a long time ago and is no longer needed. On the other hand, it could contain a heavily used variable that was initialized early and is in constant use.

Notice that, even if we select for replacement a page that is in active use, everything still works correctly. After we replace an active page with a new one, a fault occurs almost immediately to retrieve the active page. Some other page must be replaced to bring the active page back into memory. Thus, a bad replacement choice increases the page-fault rate and slows process execution. It does not, however, cause incorrect execution.

- LRU: Least recently used.

9.4.4 **<u>LRU Page Replacement</u>** lf the optimal algorithm is not feasible, perhaps an approximation of the optimal algorithm is possible. The key distinction between the FIFO and OPT algorithms (other than looking backward versus forward in time) is that the FIFO algorithm uses the time when a page was brought into memory, whereas the OPT algorithm uses the time when a page is to be used. If we use the recent past as an approximation of the near future, then we can

replace the that has not been used for the longest period of time. This approach is the LRU replacement associates with each page the time of that page's last use. When a page must be replaced, LRU chooses the page that has not been used for the longest period of time. We can think of this strategy as the optimal page-replacement algorithm looking backward in time, rather than forward. (Strangely, if we let sR be the reverse of a reference stringS, then the page-fault rate for the OPT algorithm on Sis the same as the page-fault rate for the OPT algorithm on SR. Similarly, the page-fault rate for the LRU algorithm on Sis the same as the page-fault rate for the LRU algorithm on sR.) The result of applying LRU replacement to our example reference string is shown in Figure 9.15. The LRU algorithm produces twelve faults. Notice that the first five faults are the same as those for optimal replacement. When the reference to page 4 occurs, however, LRU replacement sees that, of the three frames in memory, page 2 was used least recently. Thus, the LRU algorithm replaces page 2, not knowing that page 2 is about to be used. When it then faults for page 2, the LRU algorithm replaces page 3, since it is now the least recently used of the three pages in memory. Despite these problems, LRU replacement with twelve faults is much better than FIFO replacement with fifteen. The LRU policy is often used as a page-replacement algorithm and is considered to be good. The major problem is how to implement LRU replacement. An LRU page-replacement algorithm may require substantial hardware assistance. The problem is to determine an order for the frames defined by the time of last use. Two implementations are feasible: Counters. In the simplest case, we associate with each page-table entry a time-of-use field and add to the CPU a logical clock or counter. The clock is

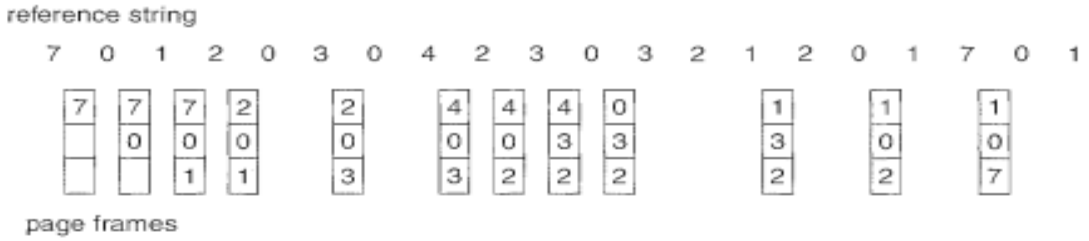

Figure 9.15 LRU page-replacement algorithm.

incremented for every memory reference. Whenever a reference to a page is made, the contents of the clock register are copied to the time-of-use field in the page-table entry for that page. In this way, we always have the "time" of the last reference to each page. We replace the page with the smallest time value. This scheme requires a search of the page table to find the LRU page and a write to memory (to the time-of-use field in the page table) for each memory access. The times must also be maintained when page tables are changed (due to CPU scheduling). Overflow of the clock must be considered.

- **Stack.** Another approach to implementing LRU replacement is to keep a stack of page numbers. Whenever a page is referenced, it is removed from the stack and put on the top. In this way, the most recently used page is always at the top of the stack and the least recently used page is always at the bottom (Figure 9.16). Because entries must be removed from the middle of the stack, it is best to implement this approach by using a doubly linked list with a head pointer and a tail pointer. Removing a page and putting it on the top of the stack then requires changing six pointers at worst. Each update is a little more expensive, but there is no search for a replacement; the tail pointer points to the bottom of the stack, which is the LRU page. This approach is particularly appropriate for software or microcode implementations of LRU replacement.

Like optimal replacement, LRU replacement does not suffer from Belady's anomaly. Both belong to a class of page-replacement algorithms, called stack algorithms, that can never exhibit Belady's anomaly. A stack algorithm is an algorithm for which it can be shown that the set of pages in memory for n frames is always a *subset* of the set of pages that would be in memory with $n + 1$ frames. For LRU replacement, the set of pages in memory would be the n most recently referenced pages. If the number of frames is increased, these n pages will still be the most recently referenced and so will still be in memory.

Note that neither implementation of LRU would be conceivable without hardware assistance beyond the standard TLB registers. The updating of the

- Optimal: Replace page not used for the longest future time.

9.4.3 Optimal Page Replacement

One result of the discovery of Belady's anomaly was the search for an optimal page-replacement algorithm, which has the lowest page-fault rate of all algorithms and will never suffer from Belady's anomaly. Such an algorithm does exist and has been called OPT or MIN. It is simply this:

> Replace the page that will not be used for the longest period of time.

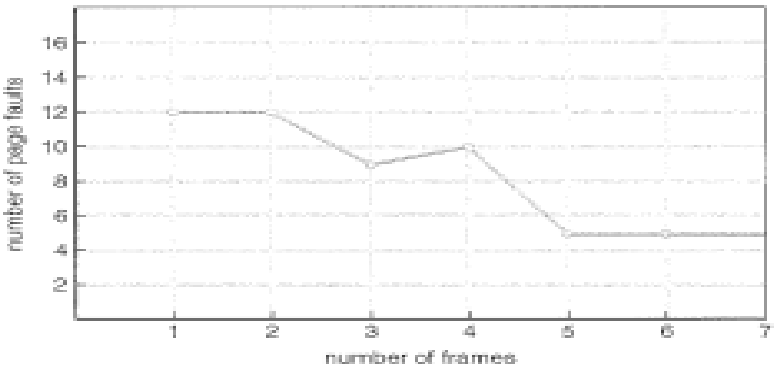

Figure 9.13 Page-fault curve for FIFO replacement on a reference string.

Use of this page-replacement algorithm guarantees the lowest possible page-fault rate for a fixed number of frames.

For example, on our sample reference string, the optimal page-replacement algorithm would yield nine page faults, as shown in Figure 9.14. The first three references cause faults that fill the three empty frames. The reference to page 2 replaces page 7, because page 7 will not be used until reference 18, whereas page 0 will be used at 5, and page 1 at 14. The reference to page 3 replaces page 1, as page 1 will be the last of the three pages in memory to be referenced again. With only nine page faults, optimal replacement is much better than a FIFO algorithm, which results in fifteen faults. (If we ignore the first three, which all algorithms must suffer, then optimal replacement is twice as good as FIFO replacement.) In fact, no replacement algorithm can process this reference string in three frames with fewer than nine faults.

Unfortunately, the optimal page-replacement algorithm is difficult to implement, because it requires future knowledge of the reference string. (We encountered a similar situation with the SJF CPU-scheduling algorithm in Section 5.3.2.) As a result, the optimal algorithm is used mainly for comparison studies. For instance, it may be useful to know that, although a new algorithm

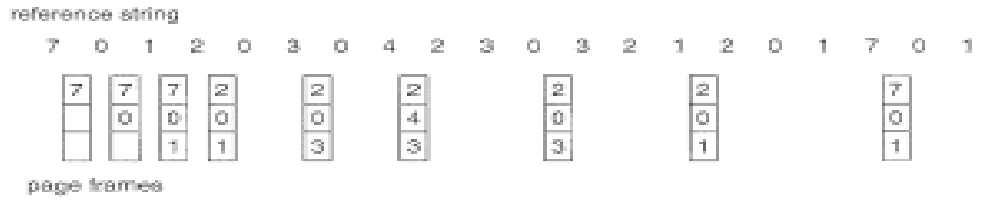

Figure 9.14 Optimal page-replacement algorithm.

- Others: Clock, NRU.

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8. Allocation of Frames

Frames allocated per process based on policies:

- Fixed allocation
- Proportional allocation (based on process size)
- Priority allocation (based on importance)

We turn next to the issue of allocation. How do we allocate the fixed amount of free memory among the various processes? If we have 93 free frames and two processes, how many frames does each process get? The simplest case is the single-user system. Consider a single-user system with 128 KB of memory composed of pages 1 KB in size. This system has 128 frames. The operating system may take 35 KB, leaving 93 frames for the user process. Under pure demand paging, all 93 frames would initially be put on the free-frame list. When a user process started execution, it would generate a sequence of page faults. The first 93 page faults would all get free frames from the free-frame list. When the free-frame list was exhausted, a page-replacement algorithm would be used to select one of the 93 in-memory pages to be replaced with the 94th, and so on. When the process terminated, the 93 frames would once again be placed on the free-frame list. There are many variations on this simple strategy. We can require that the operating system allocate all its buffer and table space from the free-frame list. When this space is not in use by the operating system, it can be used to support user paging. We can try to keep three free frames reserved on the free-frame list at all times. Thus, when a page fault occurs, there is a free frame available to page into. While the page swap is taking place, a replacement can be selected, which is then written to the disk as the user process continues to execute. Other variants are also possible, but the basic strategy is clear: the user process is allocated any free frame. 9.5.1 Minimum Number of Frames Our strategies for the allocation of frames are constrained in various ways. We cannot, for example, allocate more than the total number of available frames (unless there is page sharing). We must also allocate at least a minimum number of frames. Here, we look more closely at the latter requirement. One reason for allocating at least a minimum number of frames involves performance. Obviously, as the number of frames allocated to each process decreases, the page-fault rate increases, slowing process execution. In addition, remember that when a page fault occurs before an executing ilcstruction is complete, the instruction must be restarted. Consequently. we must have enough frames to hold all the different pages that any single ilcstruction can reference. For example, consider a machine in which all memory-reference instructions may reference only one memory address. In this case, we need at least one frame for the instruction and one frame for the mernory reference. In addition, if one-level indirect addressing is allowed (for example, a load instruction on page 16 can refer to an address on page 0, which is an indirect reference to page 23), then paging requires at least three frames per process. Think about what might happen if a process had only two frames. The minimum number of frames is defined by the computer architecture. For example, the move instruction for the PDP-11 includes more than one word for some addressing modes, and thus the ilcstruction itself may straddle two pages. In addition, each of its two operands may be indirect references, for a total of six frames. Another example is the IBM 370 MVC instruction. Since the 9.5 383 instruction is from storage location to storage location, it takes 6 bytes and can straddle two pages. The block of characters to move and the area to which it is to be m.oved can each also straddle two pages. This situation would require six frames. The worst case occurs when the MVC instruction is the operand of an

EXECUTE instruction that straddles a page boundary; in this case, we need eight frames. The worst-case scenario occurs in computer architectures that allow multiple levels of indirection (for example, each 16-bit word could contain a 15-bit address plus a 1-bit indirect indicator). Theoretically, a simple load instruction could reference an indirect address that could reference an indirect address (on another page) that could also reference an indirect address (on yet another page), and so on, until every page in virtual memory had been touched. Thus, in the worst case, the entire virtual memory must be in physical memory. To overcome this difficulty, we must place a limit on the levels of indirection (for example, limit an instruction to at most 16levels of indirection). When the first indirection occurs, a counter is set to 16; the counter is then decremented for each successive irtdirection for this instruction. If the counter is decremented to 0, a trap occurs (excessive indirection). This limitation reduces the maximum number of memory references per instruction to 17, requiring the same number of frames. Whereas the minimum number of frames per process is defined by the architecture, the maximum number is defined by the amount of available physical memory. In between, we are still left with significant choice in frame allocation.

9.5.3 Global versus Local Allocation

Another important factor in the way frames are allocated to the various processes is page replacement. With multiple processes competing for frames, we can classify page-replacement algorithms into two broad categories: global replacement and local replacement. Global replacement allows a process to select a replacement frame from the set of all frames, even if that frame is currently allocated to some other process; that is, one process can take a frame from another. Local replacement requires that each process select from only its own set of allocated frames.

For example, consider an allocation scheme wherein we allow high-priority processes to select frames from low-priority processes for replacement. A process can select a replacement from among its own frames or the frames of any lower-priority process. This approach allows a high-priority process to increase its frame allocation at the expense of a low-priority process. With a local replacement strategy, the number of frames allocated to a process does not change. With global replacement, a process may happen to select only frames allocated to other processes, thus increasing the number of frames allocated to it (assuming that other processes do not choose *its* frames for replacement).

One problem with a global replacement algorithm is that a process cannot control its own page-fault rate. The set of pages in memory for a process depends not only on the paging behavior of that process but also on the paging behavior of other processes. Therefore, the same process may perform quite

differently (for example, taking 0.5 seconds for one execution and 10.3 seconds for the next execution) because of totally external circumstances. Such is not the case with a local replacement algorithm. Under local replacement, the set of pages in memory for a process is affected by the paging behavior of only that process. Local replacement might hinder a process, however, by not making available to it other, less used pages of memory. Thus, global replacement generally results in greater system throughput and is therefore the more common method.

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9. Thrashing:-

If the number of frames allocated to a low-priority process falls below the minimum number required by the computer architecture, we must suspend that process's execution. We should then page out its remaining pages, freeing all its allocated frames. This provision introduces a swap-in, swap-out level of intermediate CPU scheduling. In fact, look at any process that does not have "enough" frames. If the process does not have the num.ber of frames it needs to

support pages in active use, it will quickly page-fault. At this point, it must replace some page. However, since all its pages are in active use, it must replace a page that will be needed again right away. Consequently, it quickly faults again, and again, and again, replacing pages that it must back in immediately. This high paging activity is called A process is thrashing if it is spending more time paging than executing.

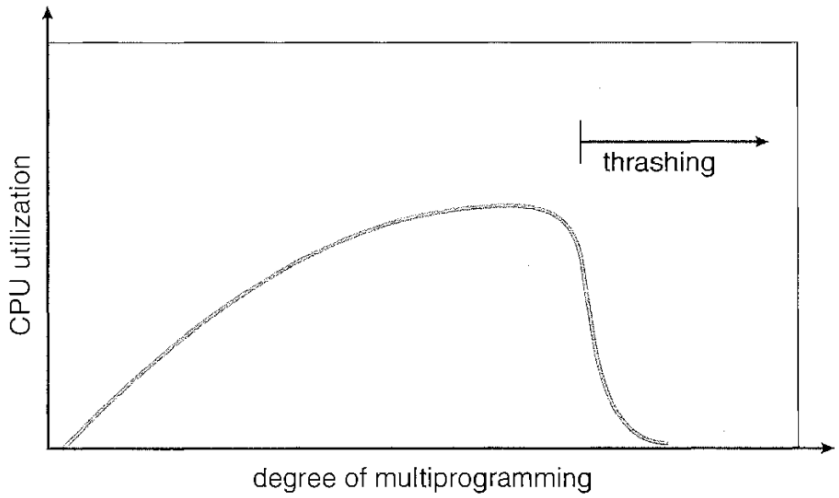

Figure 9.18 Thrashing.

thrashing, they will be in the queue for the paging device most of the time. The average service time for a page fault will increase because of the longer average queue for the paging device. Thus, the effective access time will increase even for a process that is not thrashing. To prevent thTashing, we must provide a process with as many frames as it needs. But how do we know how many frames it "needs"? There are several teclmiques. The working-set strategy (Section 9.6.2) starts by looking at how frames a process is actually using. This approach defines the locality of process execution. The locality model states that, as a process executes, it moves from locality to locality. A locality is a set of pages that are actively used together (Figure 9.19). A program is generally composed of several different localities, which may overlap. For example, when a function is called, it defines a new locality. In this locality, memory references are made to the instructions of the function call, its local variables, and a subset of the global variables. When we exit the function, the process leaves this locality, since the local variables and instructions of the function are no longer in active use. We may return to this locality later. Thus, we see that localities are defined by the program structure and its data structures. The locality model states that all programs will exhibit this basic memory reference structure. Note that the locality model is the unstated principle behind the caching discussions so far in this book If accesses to any types of data were random rather than patterned, caching would be useless. Suppose we allocate enough frames to a process to accommodate its current locality. It will fault for the pages in its locality until all these pages are in memory; then, it will not fault again until it changes localities. If we do not allocate enough frames to accommodate the size of the current locality, the process will thrash, since it cannot keep in memory all the pages that it is actively using.

Occurs when excessive paging activity slows system drastically, usually when too many processes compete for limited memory resources.

9.6.2 Working-Set Model As mentioned, the is based on the assumption of locality. This model uses a paramete1~ /':,, to define the vrindovv. The idea is to examine the most recent 6 references. The set of pages in the most recent 6 page references is the (Figure 9.20). If a page is in active use, it will be in the working set. If it is no longer being used, it will drop from the working set 6 time units after its last reference. Thus, the working set is an approximation of the program's locality. For example, given the sequence of memory references shown in Figure 9.20, if 6 = 10 memory references, then the working set at time t1 is {1, 2, 5, 6, 7}. By time t2, the working set has changed to {3, 4}. The accuracy of the working set depends on the selection of 6. If 6 is too small, it will not encompass the entire locality; if 6 is too large, it may overlap

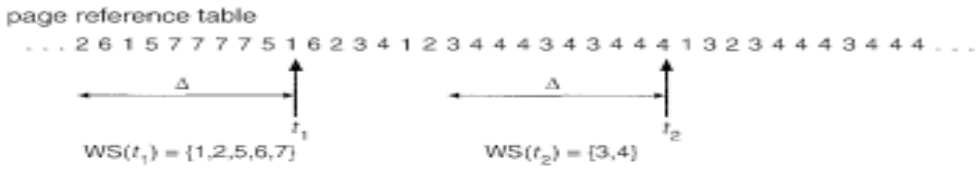

Figure 9.20 Working-set model.

several localities. In the extreme, if Δ is infinite, the working set is the set of pages touched during the process execution.

The most important property of the working set, then, is its size. If we compute the working-set size, WSS_i, for each process in the system, we can then consider that

$$D = \sum WSS_i,$$

where D is the total demand for frames. Each process is actively using the pages in its working set. Thus, process i needs WSS_i frames. If the total demand is greater than the total number of available frames ($D > m$), thrashing will occur, because some processes will not have enough frames.

Once Δ has been selected, use of the working-set model is simple. The operating system monitors the working set of each process and allocates to that working set enough frames to provide it with its working-set size. If there are enough extra frames, another process can be initiated. If the sum of the working-set sizes increases, exceeding the total number of available frames, the operating system selects a process to suspend. The process's pages are written out (swapped), and its frames are reallocated to other processes. The suspended process can be restarted later.

This working-set strategy prevents thrashing while keeping the degree of multiprogramming as high as possible. Thus, it optimizes CPU utilization.

The difficulty with the working-set model is keeping track of the working set. The working-set window is a moving window. At each memory reference, a new reference appears at one end and the oldest reference drops off the other end. A page is in the working set if it is referenced anywhere in the working-set window.

We can approximate the working-set model with a fixed-interval timer interrupt and a reference bit. For example, assume that Δ equals 10,000 references and that we can cause a timer interrupt every 5,000 references. When we get a timer interrupt, we copy and clear the reference-bit values for each page. Thus, if a page fault occurs, we can examine the current reference bit and two in-memory bits to determine whether a page was used within the last 10,000 to 15,000 references. If it was used, at least one of these bits will be on. If it has not been used, these bits will be off. Those pages with at least one bit on will be considered to be in the working set. Note that this arrangement is not entirely accurate, because we cannot tell where, within an interval of 5,000, a reference occurred. We can reduce the uncertainty by increasing the number of history bits and the frequency of interrupts (for example, 10 bits and interrupts every 1,000 references). However, the cost to service these more frequent interrupts will be correspondingly higher.

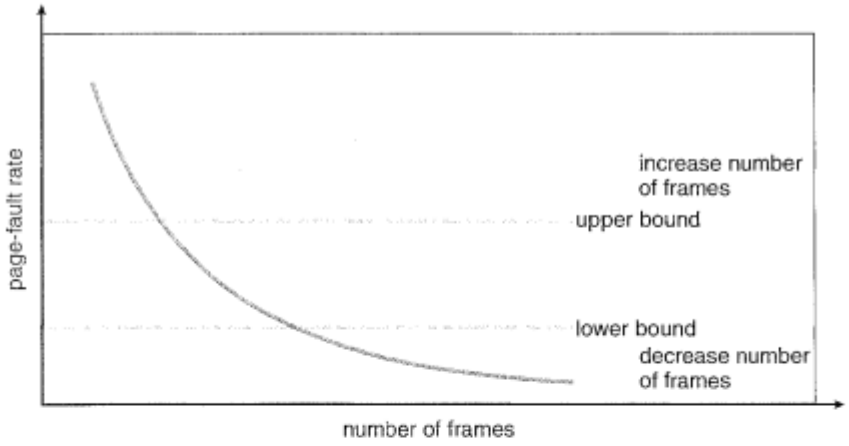

Figure 9.21 Page-fault frequency.

9.6.3 Page-Fault Frequency

The working-set model is successful, and knowledge of the working set can be useful for prepaging (Section 9.9.1), but it seems a clumsy way to control thrashing. A strategy that uses the page-fault frequency (PFF) takes a more direct approach.

The specific problem is how to prevent thrashing. Thrashing has a high page-fault rate. Thus, we want to control the page-fault rate. When it is too high, we know that the process needs more frames. Conversely, if the page-fault rate is too low, then the process may have too many frames. We can establish upper and lower bounds on the desired page-fault rate (Figure 9.21). If the actual page-fault rate exceeds the upper limit, we allocate the process another frame; if the page-fault rate falls below the lower limit, we remove a frame from the process. Thus, we can directly measure and control the page-fault rate to prevent thrashing.

As with the working-set strategy, we may have to suspend a process. If the page-fault rate increases and no free frames are available, we must select some process and suspend it. The freed frames are then distributed to processes with high page-fault rates.